

Part 1 : Social Thinking

Chapter 3: Social Beliefs and Judgments

As U.S. senators, Republican John McCain and Democrat Barack Obama each adopted positions of apparent conscience. In 2001, McCain voted against President Bush's proposed tax cut, saying "I cannot in good conscience support a tax cut in which so many of the benefits go to the most fortunate." In 2008, when McCain was campaigning for the Republican nomination and then for president, he supported and favored extending the cuts that he earlier had opposed.

Barack Obama in 2007 declared himself a "longtime advocate" of public financing of presidential elections and pledged to accept public financing should he win the Democratic nomination for president. But when he won the nomination, supported by unprecedented campaign contributions, he rejected public financing of his own campaign.

For Democrats, McCain's reversal displayed not moral courage and an openness to changing one's mind in the light of new information, but rather expedience and hypocrisy as McCain sought to pick up contributions and votes from wealthy conservatives.

For Republicans, Obama's reversal likewise displayed not a temporary strategy en route to reforming election financing, but rather hypocrisy and the same old do-what-you-can-to-get-elected politics.

As the candidates debated, most McCain partisans were impressed by the reasonableness and power of his straight-talk arguments, while being underwhelmed by the force and cogency of Obama's performance. Most Obama partisans experienced a mirror-image reaction, feeling cheered by what they perceived as their candidate's superior charisma, intelligence, and vision.

These differing reactions, which have been replicated in political perceptions across the world, illustrate the extent to which we construct social perceptions and beliefs as we

- perceive and recall events through the filters of our own assumptions;
- judge events, informed by our intuition, by implicit rules that guide our snap judgments, and by our moods;
- explain events by sometimes attributing them to the situation, sometimes to the person; and
- expect certain events, which sometimes helps bring them about.

This chapter therefore explores how we perceive, judge, and explain our social worlds, and how—and to what extent—our expectations matter.

PERCEIVING OUR SOCIAL WORLD

Striking research reveals the extent to which our assumptions and prejudgments guide our perceptions, interpretations, and recall.

Chapter 1 noted a significant fact about the human mind: our preconceptions guide how we perceive and interpret information. We construe the world through belief-tinted glasses. "Sure, preconceptions matter," people will agree; yet they fail to realize how great the effect is.

Let's consider some provocative experiments. The first group of experiments examines how predispositions and prejudgments affect how we perceive and interpret information. The second group plants a judgment in people's minds

after they have been given information to see how after-the-fact ideas bias recall. The overarching point: ***We respond not to reality as it is but to reality as we construe it.***

Priming

Unattended stimuli can subtly influence how we interpret and recall events. Imagine yourself, during an experiment, wearing earphones and concentrating on ambiguous spoken sentences such as “We stood by the bank.” When a pertinent word (river or money) is simultaneously sent to your other ear, you don’t consciously hear it. Yet the word “primes” your interpretation of the sentence (Baars & McGovern, 1994).

Our memory system is a web of associations, and ***priming*** is the awakening or activating of certain associations. Experiments show that priming one thought, even without awareness, can influence another thought, or even an action. John Bargh and his colleagues (1996) asked people to complete a sentence containing words such as “old,” “wise,” and “retired.” Shortly afterward, they observed these people walking more slowly to the elevator than did those not primed with aging-related words. Moreover, the slow walkers had no awareness of their walking speed or of having just viewed words that primed aging.

Often our thinking and acting are subtly primed by unnoticed events. Rob Holland and his colleagues (2005) observed that Dutch students exposed to the scent of an all-purpose cleaner were quicker to identify cleaning-related words. In follow-up experiments, other students exposed to a cleaning scent recalled more cleaning-related activities when describing their day’s activities and even kept their desk cleaner while eating a crumbly cookie. Moreover, all these effects occurred without the participants’ conscious awareness of the scent and its influence.

Priming experiments (Bargh, 2006) have their counterparts in everyday life:

- Watching a scary movie alone at home can activate emotions that, without our realizing it, cause us to interpret furnace noises as a possible intruder.
- Depressed moods, as this chapter explains later, prime negative associations. Put people in a good mood and suddenly their past seems more wonderful, their future brighter.
- Watching violence primes people to interpret ambiguous actions (a shove) and words (“punch”) as aggressive.
- For many psychology students, reading about psychological disorders primes how they interpret their own anxieties and gloomy moods. Reading about disease symptoms similarly primes medical students to worry about their congestion, fever, or headache.

In a host of studies, priming effects surface even when the stimuli are presented subliminally—too briefly to be perceived consciously. What’s out of sight may not be completely out of mind. An electric shock that is too slight to be felt may increase the perceived intensity of a later shock. An imperceptibly flashed word, “bread,” may prime people to detect a related word such as “butter” more quickly than they detect an unrelated word such as “bottle” or “bubble.” A subliminal color name facilitates speedier identification when the color appears on the computer screen, whereas an unseen wrong name delays color identification (Epley & others, 1999; Merikle & others, 2001). In each case, an invisible image or word primes a response to a later task.

Studies of how implanted ideas and images can prime our interpretations and recall illustrate one of this book’s take-home lessons from twenty-first-century social psychology: ***Much of our social information processing is automatic.*** It is unintentional, out of sight, and happens without our conscious awareness.

Perceiving and Interpreting Events

Despite some startling and oft-confirmed biases and logical flaws in how we perceive and understand one another, we're mostly accurate (Jussim, 2005). Our first impressions of one another are more often right than wrong. Moreover, the better we know people, the more accurately we can read their minds and feelings.

But on occasion our prejudgments err. The effects of prejudgments and expectations are standard fare for psychology's introductory course. Recall the Dalmatian photo in Chapter 1. Or consider this phrase:

A
BIRD
IN THE
THE HAND

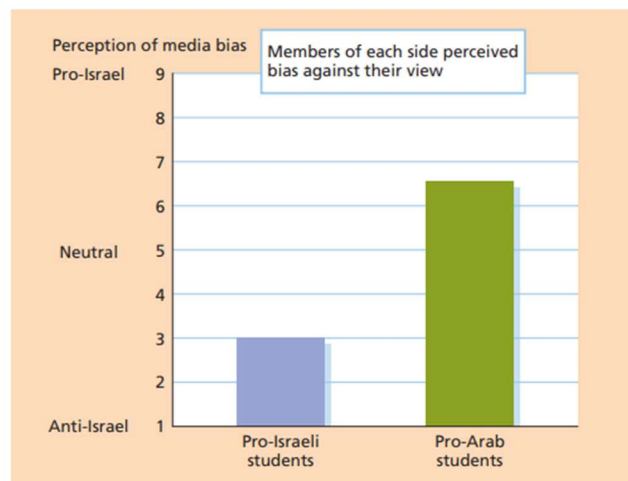
Did you notice anything wrong with it? There is more to perception than meets the eye. The same is true of social perception. Because social perceptions are very much in the eye of the beholder, even a simple stimulus may strike two people quite differently. Saying Britain's Gordon Brown is "an okay prime minister" may sound like a put-down to one of his ardent admirers and like praise to someone who regards him with contempt. When social information is subject to multiple interpretations, preconceptions matter (Hilton & von Hippel, 1990).

An experiment by Robert Vallone, Lee Ross, and Mark Lepper (1985) reveals just how powerful preconceptions can be. They showed pro-Israeli and pro-Arab students six network news segments describing the 1982 killing of civilian refugees at two camps in Lebanon. As Figure 3.1 illustrates, each group perceived the networks as hostile to its side.

FIGURE :: 3.1

Pro-Israeli and pro-Arab students who viewed network news descriptions of the "Beirut massacre" believed the coverage was biased against their point of view.

Source: Data from Vallone, Ross, & Lepper, 1985.



The phenomenon is commonplace: Sports fans perceive referees as partial to the other side. Political candidates and their supporters nearly always view the news media as unsympathetic to their cause (Richardson & others, 2008). In the 2008 U.S. presidential race, supporters of Hillary Clinton, Barack Obama, and John McCain all noted instances when the media seemed biased against their candidate, sometimes because of seeming prejudice related to gender, race, or age.

"Once you have a belief, it influences how you perceive all other relevant information. Once you see a country as hostile, you are likely to interpret ambiguous actions on their part as signifying their hostility."

—POLITICAL SCIENTIST
ROBERT JERVIS (1985)

But it's not just fans and politicians. People everywhere perceive mediators and media as biased against their position. "There is no subject about which people are less objective than objectivity," noted one media commentator (Poniewozik, 2003). Indeed, people's perceptions of bias can be used to assess their attitudes (Saucier & Miller, 2003). Tell me where you see bias, and you will signal your attitudes.

Our assumptions about the world can even make contradictory evidence seem supportive. For example, Ross and Lepper assisted Charles Lord (1979) in asking two groups of students to evaluate the results of two supposedly new research studies. Half the students favored capital punishment and half opposed it. Of the studies they evaluated, one confirmed and the other

disconfirmed the students' beliefs about the deterrent effect of the death penalty. The results: Both proponents and opponents of capital punishment readily accepted evidence that confirmed their belief but were sharply critical of disconfirming evidence. Showing the two sides an identical body of mixed evidence had not lessened their disagreement but increased it.

Is that why, in politics, religion, and science, ambiguous information often fuels conflict? Presidential debates in the United States have mostly reinforced predebate opinions. By nearly a 10-to-1 margin, those who already favored one candidate or the other perceived their candidate as having won (Kinder & Sears, 1985). Thus, report Geoffrey Munro and his colleagues (1997), people on both sides may become even more supportive of their respective candidates after viewing a presidential debate. Moreover, at the end of the Republican presidency of Ronald Reagan (during which inflation fell), only 8 percent of Democrats perceived that inflation had fallen. Republicans—47 percent of whom correctly perceived that it had—were similarly inaccurate and negative in their perceptions at the end of the Democratic Clinton presidency (Brooks, 2004). Partisanship predisposes perceptions.

In addition to these studies of people's preexisting social and political attitudes, researchers have manipulated people's preconceptions—with astonishing effects upon their interpretations and recollections.

Myron Rothbart and Pamela Birrell (1977) had University of Oregon students assess the facial expression of a man (Figure 3.2). Those told he was a Gestapo leader responsible for barbaric medical experiments on concentration camp inmates intuitively judged his expression as cruel. (Can you see that barely suppressed sneer?) Those told he was a leader in the anti-Nazi underground movement whose courage saved thousands of Jewish lives judged his facial expression as warm and kind. (Just look at those caring eyes and that almost smiling mouth.)



FIGURE :: 3.2

Judge for yourself: Is this person's expression cruel or kind? If told he was a Nazi, would your reading of his face differ?

Filmmakers control people's perceptions of emotion by manipulating the setting in which they see a face. They call this the "**Kulechov effect**," after a Russian film director who would skillfully guide viewers' inferences by manipulating their assumptions. Kulechov demonstrated the phenomenon by creating three short films that presented identical footage of the face of an actor with a neutral expression after viewers had first been shown one of three different scenes: a dead woman, a dish of soup, or a girl playing. As a result, in the first film the actor seemed sad, in the second thoughtful, and in the third happy.

Construal processes also color others' perceptions of us. When we say something good or bad about another, people spontaneously tend to associate that trait with us, report Lynda Mae, Donal Carlston, and John Skowronski (1999; Carlston & Skowronski, 2005)—a phenomenon they call **spontaneous trait transference**. If we go around talking about others being gossipy, people may then unconsciously associate "gossip" with us. Call someone a jerk and folks may later construe you as one. Describe someone as sensitive, loving, and compassionate, and you may seem more so. There is, it appears, intuitive wisdom in the childhood taunt, "***I'm rubber, you're glue; what you say bounces off me and sticks to you.***"

The bottom line: We view our social worlds through the spectacles of our beliefs, attitudes, and values. That is one reason our beliefs are so important; they shape our interpretation of everything else.

Belief Perseverance

Imagine a grandparent who decides, during an evening with a crying infant, that bottle feeding produces colicky babies: "Come to think of it, cow's milk obviously suits calves better than babies." If the infant turns out to be suffering a high

fever, will the sitter nevertheless persist in believing that bottle feeding causes colic (Ross & Anderson, 1982)? To find out, Lee Ross, Craig Anderson, and their colleagues planted a falsehood in people's minds and then tried to discredit it.

belief perseverance

Persistence of one's initial conceptions, as when the basis for one's belief is discredited but an explanation of why the belief might be true survives.

Their research reveals that it is surprisingly difficult to demolish a falsehood, once the person conjures up a rationale for it. Each experiment first implanted a belief, either by proclaiming it to be true or by showing the participants some anecdotal evidence. Then the participants were asked to explain why it is true. Finally, the researchers totally discredited the initial information by telling the participants the truth: The information was manufactured for the experiment, and half the participants in the experiment had received opposite information. Nevertheless, the new belief survived about 75 percent intact, presumably because the participants still retained their invented explanations for the belief. This phenomenon, called **belief perseverance**, shows that beliefs can grow their own legs and survive the discrediting of the evidence that inspired them.

An example: Anderson, Lepper, and Ross (1980) asked participants to decide whether individuals who take risks make good or bad firefighters. One group considered a risk-prone person who was a successful firefighter and a cautious person who was unsuccessful. The other group considered cases suggesting the opposite conclusion. After forming their theory that risk-prone people make better or worse firefighters, the participants wrote explanations for it—for example, that risk-prone people are brave or that cautious people have fewer accidents. Once each explanation was formed, it could exist independently of the information that initially created the belief. When that information was discredited, the participants still held their self-generated explanations and therefore continued to believe that risk-prone people really do make better or worse firefighters.

These experiments suggest that the more we examine our theories and explain how they might be true, the more closed we become to information that challenges our beliefs. Once we consider why an accused person might be guilty, why an offending stranger acts that way, or why a favored stock might rise in value, our explanations may survive challenges (Davies, 1997; Jelalian & Miller, 1984).

The evidence is compelling: Our beliefs and expectations powerfully affect how we mentally construct events. Usually, we benefit from our preconceptions, just as scientists benefit from creating theories that guide them in noticing and interpreting events. But the benefits sometimes entail a cost: **We become prisoners of our own thought patterns.** Thus, the supposed Martian “canals” that twentieth-century astronomers delighted in spotting turned out to be the product of intelligent life—an intelligence on Earth's side of the telescope. As another example, Germans, who widely believed that the introduction of the Euro currency led to increased prices, overestimated such price increases when comparing actual restaurant menus—the prior menu with German Mark prices and a new one with Euro prices (Traut-Mattausch & others, 2004). As an old Chinese proverb says, **“Two-thirds of what we see is behind our eyes.”**

Belief perseverance may have important consequences, as Stephan Lewandowsky and his international collaborators (2005) discovered when they explored implanted and discredited information about the Iraq war that began in 2003. As the war unfolded, the Western media reported and repeated several claims—for example, that Iraqi forces executed coalition prisoners of war—that later were shown to be false and were retracted. Alas, having accepted the information, which fit their pre-existing assumptions, Americans tended to retain the belief (unlike most Germans and Australians, who had questioned the war's rationale).

Is there a remedy for belief perseverance? There is: Explain the opposite. Charles Lord, Mark Lepper, and Elizabeth Preston (1984) repeated the capital punishment study described earlier and added two variations. First, they asked some of their participants, when evaluating the evidence, to be “as objective and unbiased as possible.” That instruction accomplished nothing; whether for or against capital punishment, those who had received the plea made evaluations as biased as those who had not.

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The researchers asked a third group to consider the opposite—to ask themselves “whether you would have made the same high or low evaluations had exactly the same study produced results on the other side of the issue.” After imagining an opposite finding, these people were much less biased in their evaluations of the evidence for and against their views. In his experiments, Craig Anderson (1982; Anderson & Sechler, 1986) consistently found that explaining why an opposite theory might be true—why a cautious rather than a risk-taking person might be a better firefighter—reduces or eliminates belief perseverance. Indeed, explaining any alternative outcome, not just the opposite, drives people to ponder various possibilities (Hirt & Markman, 1995).

Constructing Memories of Ourselves and Our Worlds

Do you agree or disagree with this statement?

Memory can be likened to a storage chest in the brain into which we deposit material and from which we can withdraw it later if needed. Occasionally, something is lost from the “chest,” and then we say we have forgotten.

About 85 percent of college students said they agreed (Lamal, 1979). As one magazine ad put it, “Science has proven the accumulated experience of a lifetime is preserved perfectly in your mind.”

Actually, psychological research has proved the opposite. Our memories are not exact copies of experiences that remain on deposit in a memory bank. Rather, ***we construct memories at the time of withdrawal***. Like a paleontologist inferring the appearance of a dinosaur from bone fragments, we reconstruct our distant past by using our current feelings and expectations to combine information fragments. Thus, we can easily (though unconsciously) revise our memories to suit our current knowledge. When one of my sons complained, “The June issue of Cricket never came,” and was then shown where it was, he delightedly responded, “Oh good, I knew I’d gotten it.”

When an experimenter or a therapist manipulates people’s presumptions about their past, a sizable percentage of people will construct false memories. Asked to imagine vividly a made-up childhood experience in which they ran, tripped, fell, and stuck their hand through a window, or knocked over a punch bowl at a wedding, about one-fourth will later recall the fictitious event as something that actually happened (Loftus & Bernstein, 2005). In its search for truth, the mind sometimes constructs a falsehood.

In experiments involving more than 20,000 people, Elizabeth Loftus (2003, 2007) and her collaborators have explored our mind’s tendency to construct memories. In the typical experiment, people witness an event, receive misleading information about it (or not), and then take a memory test. The repeated finding is the ***misinformation effect***. People incorporate the misinformation into their memories: They recall a yield sign as a stop sign, hammers as screwdrivers, Vogue magazine as Mademoiselle, Dr. Henderson as “Dr. Davidson,” breakfast cereal as eggs, and a clean-shaven man as a fellow with a mustache. Suggested misinformation may even produce false memories of supposed child sexual abuse, argues Loftus.

This process affects our recall of social as well as physical events. Jack Croxton and his colleagues (1984) had students spend 15 minutes talking with someone. Those who were later informed that this person liked them recalled the person’s behavior as relaxed, comfortable, and happy. Those informed that the person disliked them recalled the person as nervous, uncomfortable, and not so happy.

RECONSTRUCTING OUR PAST ATTITUDES

Five years ago, how did you feel about nuclear power? About your country’s president or prime minister? About your parents? If your attitudes have changed, what do you think is the extent of the change?

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Experimenters have explored such questions, and the results have been unnerving. People whose attitudes have changed often insist that they have always felt much as they now feel. Daryl Bem and Keith McConnell (1970) conducted a survey among Carnegie-Mellon University students. Buried in it was a question concerning student control over the university curriculum. A week later the students agreed to write an essay opposing student control. After doing so, their attitudes shifted toward greater opposition to student control. When asked to recall how they had answered the question before writing the essay, the students “remembered” holding the opinion that they now held and denied that the experiment had affected them.

After observing Clark University students similarly denying their former attitudes, researchers D. R. Wixon and James Laird (1976) commented, “The speed, magnitude, and certainty” with which the students revised their own histories “was striking.” As George Vaillant (1977) noted after following adults through time, “It is all too common for caterpillars to become butterflies and then to maintain that in their youth they had been little butterflies. Maturation makes liars of us all.”

The construction of positive memories brightens our recollections. Terence Mitchell, Leigh Thompson, and their colleagues (1994, 1997) report that people often exhibit rosy retrospection—they recall mildly pleasant events more favorably than they experienced them. College students on a three-week bike trip, older adults on a guided tour of Austria, and undergraduates on vacation all reported enjoying their experiences as they were having them. But they later recalled such experiences even more fondly, minimizing the unpleasant or boring aspects and remembering the high points. Thus, the pleasant times during which I have sojourned in Scotland I now (back in my office facing deadlines and interruptions) romanticize as pure bliss. The mist and the midges are but dim memories. The spectacular scenery and the fresh sea air and the favorite tea rooms are still with me. With any positive experience, some of our pleasure resides in the anticipation, some in the actual experience, and some in the rosy retrospection.

Cathy McFarland and Michael Ross (1985) found that as our relationships change, we also revise our recollections of other people. They had university students rate their steady dating partners. Two months later, they rated them again. Students who were more in love than ever had a tendency to recall love at first sight. Those who had broken up were more likely to recall having recognized the partner as somewhat selfish and bad-tempered.

Diane Holmberg and John Holmes (1994) discovered the phenomenon also operating among 373 newlywed couples, most of whom reported being very happy. When resurveyed two years later, those whose marriages had soured recalled that things had always been bad. The results are “frightening,” say Holmberg and Holmes: “Such biases can lead to a dangerous downward spiral. The worse your current view of your partner is, the worse your memories are, which only further confirms your negative attitudes.”

It’s not that we are totally unaware of how we used to feel, just that when memories are hazy, current feelings guide our recall. When widows and widowers try to recall the grief they felt on their spouse’s death five years earlier, their current emotional state colors their memories (Safer & others, 2001). When patients recall their previous day’s headache pain, their current feelings sway their recollections (Eich & others, 1985). Parents of every generation bemoan the values of the next generation, partly because they misrecall their youthful values as being closer to their current values. And teens of every generation recall their parents as—depending on their current mood—wonderful or woeful (Bornstein & others, 1991).

RECONSTRUCTING OUR PAST BEHAVIOR

Memory construction enables us to revise our own histories. The hindsight bias, described in Chapter 1, involves memory revision. Hartmut Blank and his colleagues (2003) showed this when inviting University of Leipzig students, after a surprising German election outcome, to recall their voting predictions from two months previous. The students misrecalled their predictions as closer to the actual results.

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Our memories reconstruct other sorts of past behaviors as well. Michael Ross, Cathy McFarland, and Garth Fletcher (1981) exposed some University of Waterloo students to a message convincing them of the desirability of toothbrushing. Later, in a supposedly different experiment, these students recalled brushing their teeth more often during the preceding two weeks than did students who had not heard the message. Likewise, people who are surveyed report smoking many fewer cigarettes than are actually sold (Hall, 1985). And they recall casting more votes than were actually recorded (Census Bureau, 1993).

Social psychologist Anthony Greenwald (1980) noted the similarity of such findings to happenings in George Orwell's novel 1984—in which it was “necessary to remember that events happened in the desired manner.” Indeed, argued Greenwald, we all have “totalitarian egos” that revise the past to suit our present views. Thus, ***we underreport bad behavior and overreport good behavior.***

Sometimes our present view is that we've improved—in which case we may misrecall our past as more unlike the present than it actually was. This tendency resolves a puzzling pair of consistent findings: Those who participate in psychotherapy and self-improvement programs for weight control, antismoking, and exercise show only modest improvement on average. Yet they often claim considerable benefit (Myers, 2010). Michael Conway and Michael Ross (1986) explain why: Having expended so much time, effort, and money on self-improvement, people may think, “I may not be perfect now, but I was worse before; this did me a lot of good.”

In later chapters, we will see that psychiatrists and clinical psychologists are not immune to these human tendencies. We all selectively notice, interpret, and recall events in ways that sustain our ideas. Our social judgments are a mix of observation and expectation, reason and passion.

JUDGING OUR SOCIAL WORLDS

As we have already noted, our cognitive mechanisms are efficient and adaptive, yet occasionally error-prone. Usually they serve us well. But sometimes clinicians misjudge patients, employers misjudge employees, people of one race misjudge people of another, and spouses misjudge their mates. The results can be misdiagnoses, labor strife, prejudices, and divorces. So, how—and how well—do we make intuitive social judgments?

When historians describe social psychology's first century, they will surely record 1980–2010 as the era of social cognition. By drawing on advances in cognitive psychology—in how people perceive, represent, and remember events—social psychologists have shed welcome light on how we form judgments. Let's look at what that research reveals of the marvels and mistakes of our social intuition.

Intuitive Judgments

What are our powers of intuition—of immediately knowing something without reasoning or analysis? Advocates of “intuitive management” believe we should tune into our hunches. When judging others, they say, we should plug into the nonlogical smarts of our “right brain.” When hiring, firing, and investing, we should listen to our premonitions. In making judgments, we should follow the example of Star Wars' Luke Skywalker by switching off our computer guidance systems and trusting the force within.

Are the intuitionists right that important information is immediately available apart from our conscious analysis? Or are the skeptics correct in saying that intuition is “our knowing we are right, whether we are or not”.

Priming research suggests that the unconscious indeed controls much of our behavior. As John Bargh and Tanya Chartrand (1999) explain, “Most of a person's everyday life is determined not by their conscious intentions and deliberate choices but by mental processes that are put into motion by features of the environment and that operate outside of conscious

awareness and guidance.” When the light turns red, we react and hit the brake before consciously deciding to do so. Indeed, reflect Neil Macrae and Lucy Johnston (1998), “**to be able to do just about anything at all** (e.g., driving, dating, dancing), **action initiation needs to be decoupled from the inefficient** (i.e., slow, serial, resource-consuming) **workings of the conscious mind, otherwise inaction inevitably would prevail.**”

THE POWERS OF INTUITION

“The heart has its reasons which reason does not know,” observed seventeenth-century philosopher-mathematician Blaise Pascal. Three centuries later, scientists have proved Pascal correct. **We know more than we know we know.** Studies of our unconscious information processing confirm our limited access to what’s going on in our minds (Bargh & Ferguson, 2000; Greenwald & Banaji, 1995; Strack & Deutsch, 2004). Our thinking is partly **controlled** (reflective, deliberate, and conscious) and— more than psychologists once supposed—partly **automatic** (impulsive, effortless, and without our awareness). Automatic, intuitive thinking occurs not “on-screen” but off-screen, out of sight, where reason does not go. Consider these examples of automatic thinking:

- *Schemas* are mental concepts or templates that intuitively guide our perceptions and interpretations. Whether we hear someone speaking of religious *sects* or *sex* depends not only on the word spoken but also on how we automatically interpret the sound.
- *Emotional reactions* are often nearly instantaneous, happening before there is time for deliberate thinking. One neural shortcut takes information from the eye or the ear to the brain’s sensory switchboard (the thalamus) and out to its emotional control center (the amygdala) before the thinking cortex has had any chance to intervene (LeDoux, 2002). Our ancestors who intuitively feared a sound in the bushes were usually fearing nothing. But when the sound was made by a dangerous predator they became more likely to survive to pass their genes down to us than their more deliberative cousins.
- Given sufficient *expertise*, people may intuitively know the answer to a problem. Master chess players intuitively recognize meaningful patterns that novices miss and often make their next move with only a glance at the board, as the situation cues information stored in their memory. Similarly, without knowing quite how, we recognize a friend’s voice after the first spoken word of a phone conversation.
- Faced with a decision but lacking the expertise to make an informed snap judgment, our *unconscious thinking* may guide us toward a satisfying choice. That’s what University of Amsterdam psychologist Ap Dijksterhuis and his co-workers (2006a, 2006b) discovered after showing people, for example, a dozen pieces of information about each of four potential apartments. Compared to people who made instant decisions or were given time to analyze the information, the most satisfying decisions were made by those who were distracted and unable to focus consciously on the problem. Although these findings are controversial (González-Vallejo & others, 2008; Newell & others, 2008), this much seems true: When facing a tough decision it often pays to take our time—even to sleep on it—and to await the intuitive result of our out-of-sight information processing.

Some things—facts, names, and past experiences—we remember explicitly (consciously). But other things—skills and conditioned dispositions—we remember implicitly, without consciously knowing or declaring that we know. It’s true of us all but most strikingly evident in people with brain damage who cannot form new explicit memories. One such person never could learn to recognize her physician, who would need to reintroduce himself with a handshake each day. One day the physician affixed a tack to his hand, causing the patient to jump with pain. When the physician next returned, he was still unrecognized (explicitly). But the patient, retaining an implicit memory, would not shake his hand.

Equally dramatic are the cases of *blindsight*. Having lost a portion of the visual cortex to surgery or stroke, people may be functionally blind in part of their field of vision. Shown a series of sticks in the blind field, they report seeing nothing. After

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correctly guessing whether the sticks are vertical or horizontal, the patients are astounded when told, “You got them all right.” Like the patient who “remembered” the painful handshake, these people know more than they know they know.

Consider your own taken-for-granted capacity to recognize a face. As you look at it, your brain breaks the visual information into subdimensions such as color, depth, movement, and form and works on each aspect simultaneously before reassembling the components. Finally, using automatic processing, your brain compares the perceived image with previously stored images. Voilà! Instantly and effortlessly, you recognize your grandmother. If intuition is immediately knowing something without reasoned analysis, then perceiving is intuition par excellence.

Subliminal stimuli may, as we have already noted, prime our thinking and reacting. Shown certain geometric figures for less than 0.01 second each, people may deny having seen anything more than a flash of light yet express a preference for the forms they saw.

So, many routine cognitive functions occur automatically, unintentionally, without awareness. We might remember how automatic processing helps us get through life by picturing our minds as functioning like big corporations. Our CEO—our controlled consciousness—attends to many of the most important, complex, and novel issues, while subordinates deal with routine affairs and matters requiring instant action. This delegation of resources enables us to react to many situations quickly and efficiently. The bottom line: ***Our brain knows much more than it tells us.***

THE LIMITS OF INTUITION

We have seen how automatic, intuitive thinking can “make us smart” (Gigerenzer, 2007). Elizabeth Loftus and Mark Klinger (1992) nevertheless speak for other cognitive scientists in having doubts about the brilliance of intuition. They report “a general consensus that the unconscious may not be as smart as previously believed.” For example, although subliminal stimuli can trigger a weak, fleeting response— enough to evoke a feeling if not conscious awareness—there is no evidence that commercial subliminal tapes can “reprogram your unconscious mind” for success. In fact, a significant body of evidence indicates that they can’t (Greenwald, 1992).

Social psychologists have explored not only our error-prone hindsight judgments but also our capacity for illusion—for perceptual misinterpretations, fantasies, and constructed beliefs. Michael Gazzaniga (1992, 1998, 2008) reports that patients whose brain hemispheres have been surgically separated will instantly fabricate— and believe—explanations of their own puzzling behaviors. If the patient gets up and takes a few steps after the experimenter flashes the instruction “walk” to the patient’s nonverbal right hemisphere, the verbal left hemisphere will instantly provide the patient with a plausible explanation (“I felt like getting a drink”).

Illusory thinking also appears in the vast new literature on how we take in, store, and retrieve social information. As perception researchers study visual illusions for what they reveal about our normal perceptual mechanisms, social psychologists study illusory thinking for what it reveals about normal information processing. These researchers want to give us a map of everyday social thinking, with the hazards clearly marked.

As we examine some of these efficient thinking patterns, remember this: Demonstrations of how people create counterfeit beliefs do not prove that all beliefs are counterfeit (though, to recognize counterfeiting, it helps to know how it’s done).

Overconfidence

So far we have seen that our cognitive systems process a vast amount of information efficiently and automatically. But our efficiency has a trade-off; as we interpret our experiences and construct memories, our automatic intuitions sometimes err. Usually, we are unaware of our flaws. The “intellectual conceit” evident in judgments of past knowledge (“I knew it all along”) extends to estimates of current knowledge and predictions of future behavior. We know we’ve

messed up in the past. But we have more positive expectations for our future performance in meeting deadlines, managing relationships, following an exercise routine, and so forth (Ross & Newby-Clark, 1998).

To explore this **overconfidence phenomenon**, Daniel Kahneman and Amos Tversky (1979) gave people factual statements and asked them to fill in the blanks, as in the following sentence: “I feel 98 percent certain that the air distance between New Delhi and Beijing is more than _____ miles but less than _____ miles.” Most individuals were overconfident: About 30 percent of the time, the correct answers lay outside the range they felt 98 percent confident about.

overconfidence phenomenon

The tendency to be more confident than correct—to overestimate the accuracy of one’s beliefs.

To find out whether overconfidence extends to social judgments, David Dunning and his associates (1990) created a little game show. They asked Stanford University students to guess a stranger’s answers to a series of questions, such as “Would you prepare for a difficult exam alone or with others?” and “Would you rate your lecture notes as neat or messy?” Knowing the type of question but not the actual questions, the participants first interviewed their target person about background, hobbies, academic interests, aspirations, astrological sign—anything they thought might be helpful. Then, while the targets privately answered 20 of the two-choice questions, the interviewers predicted their target’s answers and rated their own confidence in the predictions.

The air distance between New Delhi and Beijing is 2,500 miles.

The interviewers guessed right 63 percent of the time, beating chance by 13 percent. But, on average, they felt 75 percent sure of their predictions. When guessing their own roommates’ responses, they were 68 percent correct and 78 percent confident. Moreover, the most confident people were most likely to be overconfident. People also are markedly overconfident when judging whether someone is telling the truth or when estimating things such as the sexual history of their dating partner or the activity preferences of their roommates (DePaulo & others, 1997; Swann & Gill, 1997).

Ironically, **incompetence feeds overconfidence**. It takes competence to recognize what competence is, note Justin Kruger and David Dunning (1999). Students who score at the bottom on tests of grammar, humor, and logic are most prone to overestimating their gifts at such. Those who don’t know what good logic or grammar is are often unaware that they lack it. If you make a list of all the words you can form out of the letters in “psychology,” you may feel brilliant—but then stupid when a friend starts naming the ones you missed. Deanna Caputo and Dunning (2005) recreated this phenomenon in experiments, confirming that **our ignorance of our ignorance sustains our self-confidence**. Follow-up studies indicate that this “ignorance of one’s incompetence” occurs mostly on relatively easy-seeming tasks, such as forming words out of “psychology.” On really hard tasks, poor performers more often appreciate their lack of skill (Burson & others, 2006).

Ignorance of one’s incompetence helps explain David Dunning’s (2005) startling conclusion from employee assessment studies that “what others see in us . . . tends to be more highly correlated with objective outcomes than what we see in ourselves.” In one study, participants watched someone walk into a room, sit, read a weather report, and walk out (Borkenau & Liebler, 1993). Based on nothing more than that, their estimate of the person’s intelligence correlated with the person’s intelligence score about as well as did the person’s own self-estimate (.30 vs. .32)! If ignorance can beget false confidence, then—yikes!—where, we may ask, are you and I unknowingly deficient?

In Chapter 2 we noted that people overestimate their long-term emotional responses to good and bad happenings. Are people better at predicting their own behavior? To find out, Robert Vallone and his colleagues (1990) had college students predict in September whether they would drop a course, declare a major, elect to live off campus next year, and so forth. Although the students felt, on average, 84 percent sure of those self-predictions, they were wrong nearly twice as often as they expected to be. Even when feeling 100 percent sure of their predictions, they erred 15 percent of the time.

In estimating their chances for success on a task, such as a major exam, people’s confidence runs highest when the moment of truth is off in the future. By exam day, the possibility of failure looms larger and confidence typically drops (Gilovich & others, 1993; Shepperd & others, 2005). Roger Buehler and his colleagues (1994, 2002, 2003, 2005) report that

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most students also confidently underestimate how long it will take them to complete papers and other major assignments. They are not alone:

- *The “planning fallacy.”* How much free time do you have today? How much free time do you expect you will have a month from today? Most of us overestimate how much we’ll be getting done, and therefore how much free time we will have (Zauberman & Lynch, 2005). Professional planners, too, routinely underestimate the time and expense of projects. In 1969, Montreal Mayor Jean Drapeau proudly announced that a \$120 million stadium with a retractable roof would be built for the 1976 Olympics. The roof was completed in 1989 and cost \$120 million by itself. In 1985, officials estimated that Boston’s “Big Dig” highway project would cost \$2.6 billion and take until 1998. The cost ballooned to \$14.6 billion and the project took until 2006.
- *Stockbroker overconfidence.* Investment experts market their services with the confident presumption that they can beat the stock market average, forgetting that for every stockbroker or buyer saying “Sell!” at a given price, there is another saying “Buy!” A stock’s price is the balance point between those mutually confident judgments. Thus, incredible as it may seem, economist Burton Malkiel (2007) reports that mutual fund portfolios selected by investment analysts have not outperformed randomly selected stocks.
- *Political overconfidence.* Overconfident decision makers can wreak havoc. It was a confident Adolf Hitler who from 1939 to 1945 waged war against the rest of Europe. It was a confident Lyndon Johnson who in the 1960s invested U.S. weapons and soldiers in the effort to salvage democracy in South Vietnam. It was a confident Saddam Hussein who in 1990 marched his army into Kuwait and in 2003 promised to defeat invading armies. It was a confident George W. Bush who proclaimed that peaceful democracy would soon prevail in a liberated and thriving Iraq, with its alleged weapons of mass destruction newly destroyed.

What produces overconfidence? Why doesn’t experience lead us to a more realistic self-appraisal? For one thing, people tend to recall their mistaken judgments as times when they were almost right. Philip Tetlock (1998, 1999, 2005) observed this after inviting various academic and government experts to project—from their viewpoint in the late 1980s—the future governance of the Soviet Union, South Africa, and Canada. Five years later communism had collapsed, South Africa had become a multiracial democracy, and Canada’s French-speaking minority had not seceded. Experts who had felt more than 80 percent confident were right in predicting these turns of events less than 40 percent of the time. Yet, reflecting on their judgments, those who erred believed they were still basically right. I was “almost right,” said many. “The hardliners almost succeeded in their coup attempt against Gorbachev.” “The Quebecois separatists almost won the secessionist referendum.” “But for the coincidence of de Klerk and Mandela, there would have been a much bloodier transition to black majority rule in South Africa.” The Iraq war was a good idea, just badly executed, excused many of those who had supported it. Among political experts—and also stock market forecasters, mental health workers, and sports prognosticators—overconfidence is hard to dislodge.

CONFIRMATION BIAS

People also tend not to seek information that might disprove what they believe. P. C. Wason (1960) demonstrated this, as you can, by giving participants a sequence of three numbers—2, 4, 6—that conformed to a rule he had in mind. (The rule was simply any three ascending numbers.) To enable the participants to discover the rule, Wason invited each person to generate additional sets of three numbers. Each time, Wason told the person whether or not the set conformed to his rule. As soon as participants were sure they had discovered the rule, they were to stop and announce it.

The result? Seldom right but never in doubt: 23 of the 29 participants convinced themselves of a wrong rule. They typically formed some erroneous belief about the rule (for example, counting by twos) and then searched for *confirming* evidence (for example, by testing 8, 10, 12) rather than attempting to *disconfirm* their hunches. We are eager to verify our beliefs but less inclined to seek evidence that might disprove them, a phenomenon called the **confirmation bias**.

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Confirmation bias helps explain why our self-images are so remarkably stable. In experiments at the University of Texas at Austin, William Swann and Stephen Read (1981; Swann & others, 1992a, 1992b, 2007) discovered that students seek, elicit, and recall feedback that confirms their beliefs about themselves. People seek as friends and spouses those who bolster their own self views—even if they think poorly of themselves (Swann & others, 1991, 2003).

Swann and Read (1981) liken this *self-verification* to how someone with a domineering self-image might behave at a party. Upon arriving, the person seeks those guests whom she knows will acknowledge her dominance. In conversation she then presents her views in ways that elicit the respect she expects. After the party, she has trouble recalling conversations in which her influence was minimal and more easily recalls her persuasiveness in the conversations that she dominated. Thus, her experience at the party confirms her self-image.

REMEDIES FOR OVERCONFIDENCE

What lessons can we draw from research on overconfidence? One lesson is to be wary of other people's dogmatic statements. Even when people are sure they are right, they may be wrong. **Confidence and competence need not coincide.**

Three techniques have successfully reduced the overconfidence bias. One is **prompt feedback** (Lichtenstein & Fischhoff, 1980). In everyday life, weather forecasters and those who set the odds in horse racing both receive clear, daily feedback. And experts in both groups do quite well at estimating their probable accuracy (Fischhoff, 1982).

To reduce “planning fallacy” overconfidence, people can be asked to **unpack a task** —to break it down into its subcomponents—and estimate the time required for each. Justin Kruger and Matt Evans (2004) report that doing so leads to more realistic estimates of completion time.

When people think about why an idea might be true, it begins to seem true (Koehler, 1991). Thus, a third way to reduce overconfidence is to **get people to think of one good reason why their judgments might be wrong**; that is, **force them to consider disconfirming information** (Koriat & others, 1980). Managers might foster more realistic judgments by insisting that all proposals and recommendations include reasons why they might not work.

Still, we should be careful not to undermine people's reasonable self-confidence or to destroy their decisiveness. In times when their wisdom is needed, those lacking self-confidence may shrink from speaking up or making tough decisions. Overconfidence can cost us, but **realistic self-confidence is adaptive.**

Heuristics: Mental Shortcuts

With precious little time to process so much information, our cognitive system is fast and frugal. It specializes in mental shortcuts. With remarkable ease, we form impressions, make judgments, and invent explanations. We do so by using heuristics —simple, efficient thinking strategies. Heuristics enable us to live and make routine decisions with minimal effort (Shah & Oppenheimer, 2008). In most situations, our snap generalizations—“That's dangerous!”—are adaptive. The speed of these intuitive guides promotes our survival. The biological purpose of thinking is less to make us right than to keep us alive. In some situations, however, haste makes error.

THE REPRESENTATIVENESS HEURISTIC

University of Oregon students were told that a panel of psychologists interviewed a sample of 30 engineers and 70 lawyers and summarized their impressions in thumbnail descriptions. The following description, they were told, was drawn at random from the sample of 30 engineers and 70 lawyers:

Twice divorced, Frank spends most of his free time hanging around the country club. His clubhouse bar conversations often center around his regrets at having tried to follow his esteemed father's footsteps. The long

hours he had spent at academic drudgery would have been better invested in learning how to be less quarrelsome in his relations with other people.

Question: What is the probability that Frank is a lawyer rather than an engineer?

Asked to guess Frank’s occupation, more than 80 percent of the students surmised he was one of the lawyers (Fischhoff & Bar-Hillel, 1984). Fair enough. But how do you suppose those estimates changed when the sample description was given to another group of students, modified to say that 70 percent were engineers? Not in the slightest. The students took no account of the base rate of engineers and lawyers; in their minds Frank was more representative of lawyers, and that was all that seemed to matter.

To judge something by intuitively comparing it to our mental representation of a category is to use the **representativeness heuristic**. Representativeness (typicalness) usually is a reasonable guide to reality. But, as we saw with “Frank” above, it doesn’t always work. Consider Linda, who is 31, single, outspoken, and very bright. She majored in philosophy in college. As a student she was deeply concerned with discrimination and other social issues, and she participated in antinuclear demonstrations. Based on that description, would you say it is more likely that

- a. Linda is a bank teller.
- b. Linda is a bank teller and active in the feminist movement.

Most people think **b** is more likely, partly because Linda better represents their image of feminists (Mellers & others, 2001). But ask yourself: Is there a better chance that Linda is both a bank teller and a feminist than that she’s a bank teller (whether feminist or not)? As Amos Tversky and Daniel Kahneman (1983) reminded us, the conjunction of two events cannot be more likely than either one of the events alone.

THE AVAILABILITY HEURISTIC

Consider the following: Do more people live in Iraq or in Tanzania?

You probably answered according to how readily Iraqis and Tanzanians come to mind. If examples are readily available in our memory—as Iraqis tend to be—then we presume that other such examples are commonplace. Usually this is true, so we are often well served by this cognitive rule, called the availability heuristic (Table 3.1). Said simply, the more easily we recall something, the more likely it seems.

TABLE :: 3.1 Fast and Frugal Heuristics

Heuristic	Definition	Example	But May Lead to
Representativeness	Snap judgments of whether someone or something fits a category	Deciding that Carlos is a librarian rather than a trucker because he better represents one’s image of librarians	Discounting other important information
Availability	Quick judgments of likelihood of events (how available in memory)	Estimating teen violence after school shootings	Overweighting vivid instances and thus, for example, fearing the wrong things

But sometimes the rule deludes us. If people hear a list of famous people of one sex (Jennifer Lopez, Venus Williams, Hillary Clinton) intermixed with an equal-size list of unfamous people of the other sex (Donald Scarr, William Wood, Mel Jasper), the famous names will later be more cognitively available. Most people will subsequently recall having heard more (in this instance) women’s names (McKelvie, 1995, 1997; Tversky & Kahneman, 1973). Vivid, easy-to-imagine events, such

as shark attacks or diseases with easy-to-picture symptoms, may likewise seem more likely to occur than harder-to-picture events (MacLeod & Campbell, 1992; Sherman & others, 1985).

Even fictional happenings in novels, television, and movies leave images that later penetrate our judgments (Gerrig & Prentice, 1991; Green & others, 2002; Mar & Oatley, 2008). The more absorbed and “transported” the reader (“I could easily picture the events”), the more the story affects the reader’s later beliefs (Diekmann & others, 2000). Readers who are captivated by romance novels, for example, may gain readily available sexual scripts that influence their own sexual attitudes and behaviors.

Our use of the availability heuristic highlights a basic principle of social thinking: People are slow to deduce particular instances from a general truth, but they are remarkably quick to infer general truth from a vivid instance. No wonder that after hearing and reading stories of rapes, robberies, and beatings, 9 out of 10 Canadians overestimated—usually by a considerable margin—the percentage of crimes that involved violence (Doob & Roberts, 1988). And no wonder that South Africans, after a series of headline-grabbing gangland robberies and slayings, estimated that violent crime had almost doubled between 1998 and 2004, when actually it had decreased substantially (Wines, 2005).

The availability heuristic explains why powerful anecdotes can nevertheless be more compelling than statistical information and why perceived risk is therefore often badly out of joint with real risks (Allison & others, 1992). We fret over swine flu (H1N1) but don’t bother to get vaccinated for common flu, which kills tens of thousands. We fret over extremely rare child abduction, even if we don’t buckle our children in the backseat. We fear terrorism, but are indifferent to global climate change—“Armageddon in slow motion.” In short, we worry about remote possibilities while ignoring higher probabilities, a phenomenon that Cass Sunstein (2007b) calls our “**probability neglect**.”

Because news footage of airplane crashes is a readily available memory for most of us—especially since September 11, 2001—we often suppose we are more at risk traveling in commercial airplanes than in cars. Actually, from 2003 to 2005, U.S. travelers were 230 times more likely to die in a car crash than on a commercial flight covering the same distance (National Safety Council, 2008). In 2006, reports the Flight Safety Foundation, there was one airliner accident for every 4.2 million flights by Western-built commercial jets (Wald, 2008). For most air travelers, the most dangerous part of the journey is the drive to the airport.

Shortly after 9/11, as many people abandoned air travel and took to the roads, I estimated that if Americans flew 20 percent less and instead drove those unflown miles, we could expect an additional 800 traffic deaths in the ensuing year (Myers, 2001). It took a curious German researcher (why didn’t I think of this?) to check that prediction against accident data, which confirmed an excess of some 350 deaths in the last three months of 2001 compared with the three-month average in the preceding five years (Gigerenzer, 2004). The 9/11 terrorists appear to have killed more people unnoticed—on America’s roads—than they did with the 266 fatalities on those four planes.

By now it is clear that our naive statistical intuitions, and our resulting fears, are driven not by calculation and reason but by emotions attuned to the availability heuristic. After this book is published, there likely will be another dramatic natural or terrorist event, which will again propel our fears, vigilance, and resources in a new direction. Terrorists, aided by the media, may again achieve their objective of capturing our attention, draining our resources, and distracting us from the mundane, undramatic, insidious risks that, over time, devastate lives, such as the rotavirus that each day claims the equivalent of four 747s filled with children (Parashar & others, 2006). But then again, dramatic events can also serve to awaken us to real risks. That, say some scientists, is what happened when hurricanes Katrina and Rita in 2005 began to raise concern that global warming, by raising sea levels and spawning extreme weather, is destined to become nature’s own weapon of mass destruction.

Counterfactual Thinking

Easily imagined (cognitively available) events also influence our experiences of guilt, regret, frustration, and relief. If our team loses (or wins) a big game by one point, we can easily imagine how the game might have gone the other way, and thus we feel regret (or relief). Imagining worse alternatives helps us feel better. Imagining better alternatives, and pondering what we might do differently next time, helps us prepare to do better in the future (Epstude & Roese, 2008).

In Olympic competition, athletes' emotions after an event reflect mostly how they did relative to expectations, but also their **counterfactual thinking**—their mentally simulating what might have been (McGraw & others, 2005; Medvec & others, 1995). Bronze medalists (for whom an easily imagined alternative was finishing without a medal) exhibited more joy than silver medalists (who could more easily imagine having won the gold). On the medal stand, it has been said, happiness is as simple as 1-3-2. Similarly, the higher a student's score within a grade category (such as B+), the worse they feel (Medvec & Savitsky, 1997). The B+ student who misses an A- by a point feels worse than the B+ student who actually did worse and just made a B+ by a point.

Such counterfactual thinking occurs when we can easily picture an alternative outcome (Kahneman & Miller, 1986; Markman & McMullen, 2003):

- If we barely miss a plane or a bus, we imagine making it if only we had left at our usual time, taken our usual route, not paused to talk. If we miss our connection by a half hour or after taking our usual route, it's harder to simulate a different outcome, so we feel less frustration.
- If we change an exam answer, then get it wrong, we will inevitably think "If only . . ." and will vow next time to trust our immediate intuition—although, contrary to student lore, answer changes are more often from incorrect to correct (Kruger & others, 2005).
- The team or the political candidate that barely loses will simulate over and over how they could have won (Sanna & others, 2003).

Counterfactual thinking underlies our feelings of luck. When we have barely escaped a bad event—avoiding defeat with a last-minute goal or standing nearest a falling icicle—we easily imagine a negative counterfactual (losing, being hit) and therefore feel "good luck" (Teigen & others, 1999). "Bad luck" refers to bad events that did happen but easily might not have.

The more significant the event, the more intense the counterfactual thinking (Roese & Hur, 1997). Bereaved people who have lost a spouse or a child in a vehicle accident, or a child to sudden infant death syndrome, commonly report replaying and undoing the event (Davis & others, 1995, 1996). One friend of mine survived a head-on collision with a drunk driver that killed his wife, daughter, and mother. He recalled, "For months I turned the events of that day over and over in my mind. I kept reliving the day, changing the order of events so that the accident wouldn't occur" (Sittser, 1994).

Across Asian and Western cultures most people, however, live with less regret over things done than over things they failed to do, such as, "I wish I had been more serious in college" or "I should have told my father I loved him before he died" (Gilovich & Medvec, 1994; Rajagopal & others, 2006). In one survey of adults, the most common regret was not taking their education more seriously (Kinnier & Metha, 1989). Would we live with less regret if we dared more often to reach beyond our comfort zone—to venture out, risking failure, but at least having tried?

Illusory Thinking

Another influence on everyday thinking is our search for order in random events, a tendency that can lead us down all sorts of wrong paths.

ILLUSORY CORRELATION

It's easy to see a correlation where none exists. When we expect to find significant relationships, we easily associate random events, perceiving an **illusory correlation**. William Ward and Herbert Jenkins (1965) showed people the results of a hypothetical 50-day cloud-seeding experiment. They told participants which of the 50 days the clouds had been seeded and which days it rained. That information was nothing more than a random mix of results: Sometimes it rained after seeding; sometimes it didn't. Participants nevertheless became convinced—in conformity with their ideas about the effects of cloud seeding—that they really had observed a relationship between cloud seeding and rain.

Other experiments confirm that people easily misperceive random events as confirming their beliefs (Crocker, 1981; Jennings & others, 1982; Troler & Hamilton, 1986). If we believe a correlation exists, we are more likely to notice and recall confirming instances. If we believe that premonitions correlate with events, we notice and remember the joint occurrence of the premonition and the event's later occurrence. If we believe that overweight women are unhappier, we perceive that we have witnessed such a correlation even when we have not (Viken & others, 2005). We seldom notice or remember all the times unusual events do not coincide. If, after we think about a friend, the friend calls us, we notice and remember that coincidence. We don't notice all the times we think of a friend without any ensuing call or receive a call from a friend about whom we've not been thinking.

ILLUSION OF CONTROL

Our tendency to perceive random events as related feeds an **illusion of control**—the idea that chance events are subject to our influence. This keeps gamblers going and makes the rest of us do all sorts of unlikely things.

GAMBLING Ellen Langer (1977) demonstrated the illusion of control with experiments on gambling. Compared with those given an assigned lottery number, people who chose their own number demanded four times as much money when asked if they would sell their ticket. When playing a game of chance against an awkward and nervous person, they bet significantly more than when playing against a dapper, confident opponent. Being the person who throws the dice or spins the wheel increases people's confidence (Wohl & Enzle, 2002). In these and other ways, more than 50 experiments have consistently found people acting as if they can predict or control chance events (Presson & Benassi, 1996; Thompson & others, 1998).

Observations of real-life gamblers confirm these experimental findings. Dice players may throw softly for low numbers and hard for high numbers (Henslin, 1967). The gambling industry thrives on gamblers' illusions. Gamblers attribute wins to their skill and foresight. Losses become "near misses" or "flukes," or for the sports gambler, a bad call by the referee or a freakish bounce of the ball (Gilovich & Douglas, 1986).

Stock traders also like the "feeling of empowerment" that comes from being able to choose and control their own stock trades, as if their being in control can enable them to outperform the market average. One ad declared that online investing "is about control." Alas, the illusion of control breeds overconfidence and frequent losses after stock market trading costs are subtracted (Barber & Odean, 2001).

REGRESSION TOWARD THE AVERAGE Tversky and Kahneman (1974) noted another way by which an illusion of control may arise: We fail to recognize the statistical phenomenon of **regression toward the average**. Because exam scores fluctuate partly by chance, most students who get extremely high scores on an exam will get lower scores on the next

exam. If their first score is at the ceiling, their second score is more likely to fall back (“regress”) toward their own average than to push the ceiling even higher. That is why a student who does consistently good work, even if never the best, will sometimes end a course at the top of the class. Conversely, the lowest-scoring students on the first exam are likely to improve. If those who scored lowest go for tutoring after the first exam, the tutors are likely to feel effective when the student improves, even if the tutoring had no effect.

Indeed, when things reach a low point, we will try anything, and whatever we try—going to a psychotherapist, starting a new diet-exercise plan, reading a self-help book—is more likely to be followed by improvement than by further deterioration. Sometimes we recognize that events are not likely to continue at an unusually good or bad extreme. Experience has taught us that when everything is going great, something will go wrong, and that when life is dealing us terrible blows, we can usually look forward to things getting better. Often, though, we fail to recognize this regression effect. We puzzle at why baseball’s rookie of the year often has a more ordinary second year—did he become overconfident? Self-conscious? We forget that ***exceptional performance tends to regress toward normality***.

By simulating the consequences of using praise and punishment, Paul Schaffner (1985) showed how the illusion of control might infiltrate human relations. He invited Bowdoin College students to train an imaginary fourth-grade boy, “Harold,” to come to school by 8:30 each morning. For each school day of a three-week period, a computer displayed Harold’s arrival time, which was always between 8:20 and 8:40. The students would then select a response to Harold, ranging from strong praise to strong reprimand. As you might expect, they usually praised Harold when he arrived before 8:30 and reprimanded him when he arrived after 8:30. Because Schaffner had programmed the computer to display a random sequence of arrival times, Harold’s arrival time tended to improve (to regress toward 8:30) after he was reprimanded. For example, if Harold arrived at 8:39, he was almost sure to be reprimanded, and his randomly selected next-day arrival time was likely to be earlier than 8:39. Thus, even though their reprimands were having no effect, most students ended the experiment believing that their reprimands had been effective.

This experiment demonstrates Tversky and Kahneman’s provocative conclusion: Nature operates in such a way that we often feel punished for rewarding others and rewarded for punishing them. In actuality, as every student of psychology knows, positive reinforcement for doing things right is usually more effective and has fewer negative side effects.

Moods and Judgments

Social judgment involves efficient, though fallible, information processing. It also involves our feelings: Our moods infuse our judgments. We are not cool computing machines; we are emotional creatures. The extent to which feeling infuses cognition appears in new studies comparing happy and sad individuals (Myers, 1993, 2000b). Unhappy people—especially those bereaved or depressed—tend to be more self-focused and brooding. A depressed mood motivates intense thinking—a search for information that makes one’s environment more understandable and controllable (Weary & Edwards, 1994).

Happy people, by contrast, are more trusting, more loving, more responsive. If people are made temporarily happy by receiving a small gift while mall-shopping, they will report, a few moments later on an unrelated survey, that their cars and TV sets are working beautifully—better, if you took their word for it, than those belonging to folks who replied after not receiving gifts.

Moods pervade our thinking. To West Germans enjoying their team’s World Cup soccer victory (Schwarz & others, 1987) and to Australians emerging from a heartwarming movie (Forgas & Moylan, 1987), people seem good-hearted, life seems wonderful. After (but not before) a 1990 football game between rivals Alabama and Auburn, victorious Alabama fans deemed war less likely and potentially devastating than did the gloomier Auburn fans (Schweitzer & others, 1992). ***When we are in a happy mood, the world seems friendlier, decisions are easier, good news more readily comes to mind*** (DeSteno & others, 2000; Isen & Means, 1983; Stone & Glass, 1986).

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Let a mood turn gloomy, however, and thoughts switch onto a different track. Off come the rose-colored glasses; on come the dark glasses. Now the bad mood primes our recollections of negative events (Bower, 1987; Johnson & Magaro, 1987). Our relationships seem to sour. Our self-images take a dive. Our hopes for the future dim. And other people's behavior seems more sinister (Brown & Taylor, 1986; Mayer & Salovey, 1987).

University of New South Wales social psychologist Joseph Forgas (1999, 2008) had often been struck by how moody people's "memories and judgments change with the color of their mood." To understand this "mood infusion" he began to experiment. Imagine yourself in one such study. Using hypnosis, Forgas and his colleagues (1984) put you in a good or a bad mood and then have you watch a videotape (made the day before) of yourself talking with someone. If made to feel happy, you feel pleased with what you see, and you are able to detect many instances of your poise, interest, and social skill. If you've been put in a bad mood, viewing the same tape seems to reveal a quite different you—one who is stiff, nervous, and inarticulate (Figure 3.3). Given how your mood colors your judgments, you feel relieved at how things brighten when the experimenter switches you to a happy mood before leaving the experiment. Curiously, note Michael Ross and Garth Fletcher (1985), we don't attribute our changing perceptions to our mood shifts. Rather, the world really seems different.

Our moods color how we judge our worlds partly by bringing to mind past experiences associated with the mood. When we are in a bad mood, we have more depressing thoughts. Mood-related thoughts may distract us from complex thinking about something else. Thus, when emotionally aroused—when angry or even in a very good mood—we become more likely to make snap judgments and evaluate others based on stereotypes (Bodenhausen & others, 1994; Paulhus & Lim, 1994).

EXPLAINING OUR SOCIAL WORLDS

People make it their business to explain other people, and social psychologists make it their business to explain people's explanations. So, how—and how accurately—do people explain others' behavior? Attribution theory suggests some answers.

Our judgments of people depend on how we explain their behavior. Depending on our explanation, we may judge killing as murder, manslaughter, self-defense, or heroism. Depending on our explanation, we may view a homeless person as lacking initiative or as victimized by job and welfare cutbacks. Depending on our explanation, we may interpret someone's friendly behavior as genuine warmth or as ingratiation.

Attributing Causality: To the Person or the Situation

We endlessly analyze and discuss why things happen as they do, especially when we experience something negative or unexpected (Bohner & others, 1988; Weiner, 1985). If worker productivity declines, do we assume the workers are getting lazier? Or has their workplace become less efficient? Does a young boy who hits his classmates have a hostile personality? Or is he responding to relentless teasing? Amy Holtzworth-Munroe and Neil Jacobson (1985, 1988) found that married people often analyze their partners' behaviors, especially their negative behaviors. Cold hostility, more than a warm hug, is likely to leave the partner wondering why?

Spouses' answers correlate with marriage satisfaction. Unhappy couples usually offer distress-maintaining explanations for negative acts ("she was late because she doesn't care about me"). Happy couples more often externalize ("she was late because of heavy traffic"). With positive partner behavior, their explanations similarly work either to maintain distress ("he brought me flowers because he wants sex") or to enhance the relationship ("he brought me flowers to show he loves me") (Hewstone & Fincham, 1996; McNulty & others, 2008; Weiner, 1995).

Antonia Abbey (1987, 1991; Abbey & Ross, 1998) and her colleagues have repeatedly found that men are more likely than women to attribute a woman's friendliness to mild sexual interest. That misreading of warmth as a sexual come-on—an example of **misattribution**—can contribute to behavior that women regard as sexual harassment or even rape (Farris & others, 2008; Kolivas & Gross, 2007; Pryor & others, 1997). Many men believe women are flattered by repeated requests for dates, which women more often view as harassment (Rotundo & others, 2001).

Misattribution is particularly likely when men are in positions of power. A manager may misinterpret a subordinate woman's submissive or friendly behavior and, full of himself, see her in sexual terms (Bargh & Raymond, 1995). Men more often than women think about sex. Men also are more likely than women to assume that others share their feelings (recall from Chapter 2 the “false consensus effect”). Thus, a man may greatly overestimate the sexual significance of a woman's courtesy smile (Levesque & others, 2006; Nelson & LeBoeuf, 2002).

Such misattributions help explain the greater sexual assertiveness exhibited by men across the world and the greater tendency of men in various cultures, from Boston to Bombay, to justify rape by arguing that the victim consented or implied consent (Kanekar & Nazareth, 1988; Muehlenhard, 1988; Shotland, 1989). Women more often judge forcible sex as meriting conviction and a stiff sentence (Schutte & Hosch, 1997). Misattributions also help explain why the 23 percent of American women who say they have been forced into unwanted sexual behavior is eight times greater than the 3 percent of American men who say they have ever forced a woman into a sexual act (Laumann & others, 1994).

Attribution theory analyzes how we explain people's behavior. The variations of attribution theory share some common assumptions. As Daniel Gilbert and Patrick Malone (1995) explain, each “construes the human skin as a special boundary that separates one set of ‘causal forces’ from another. On the sunny side of the epidermis are the external or situational forces that press inward upon the person, and on the meaty side are the internal or personal forces that exert pressure outward. Sometimes these forces press in conjunction, sometimes in opposition, and their dynamic interplay manifests itself as observable behavior.”

Attribution theory pioneer Fritz Heider (1958) and others after him analyzed the “commonsense psychology” by which people explain everyday events. They concluded that when we observe someone acting intentionally, we sometimes attribute that person's behavior to internal causes (for example, the person's disposition) and sometimes to external causes (for example, something about the person's situation). A teacher may wonder whether a child's underachievement is due to lack of motivation and ability (a **dispositional attribution**) or to physical and social circumstances (a **situational attribution**). Also, some of us are more inclined to attribute behavior to stable personality; others tend more to attribute behavior to situations (Bastian & Haslam, 2006; Robins & others, 2004).

INFERRING TRAITS

Edward Jones and Keith Davis (1965) noted that we often infer that other people's actions are indicative of their intentions and dispositions. If I observe Rick making a sarcastic comment to Linda, I infer that Rick is a hostile person. Jones and Davis's “theory of correspondent inferences” specified the conditions under which people infer traits. For example, normal or expected behavior tells us less about the person than does unusual behavior. If Samantha is sarcastic in a job interview, where a person would normally be pleasant, that tells us more about Samantha than if she is sarcastic with her siblings.

The ease with which we infer traits—a phenomenon called **spontaneous trait inference**—is remarkable. In experiments at New York University, James Uleman (1989; Uleman & others, 2008) gave students statements to remember, such as “The librarian carries the old woman's groceries across the street.” The students would instantly, unintentionally, and unconsciously infer a trait. When later they were helped to recall the sentence, the most valuable clue word was not “books” (to cue librarian) or “bags” (to cue groceries) but “helpful”—the inferred trait that I suspect you, too, spontaneously attributed to the librarian. Given even just 1/10th of a second exposure to someone's face, people will spontaneously infer some personality traits (Willis & Todorov, 2006).

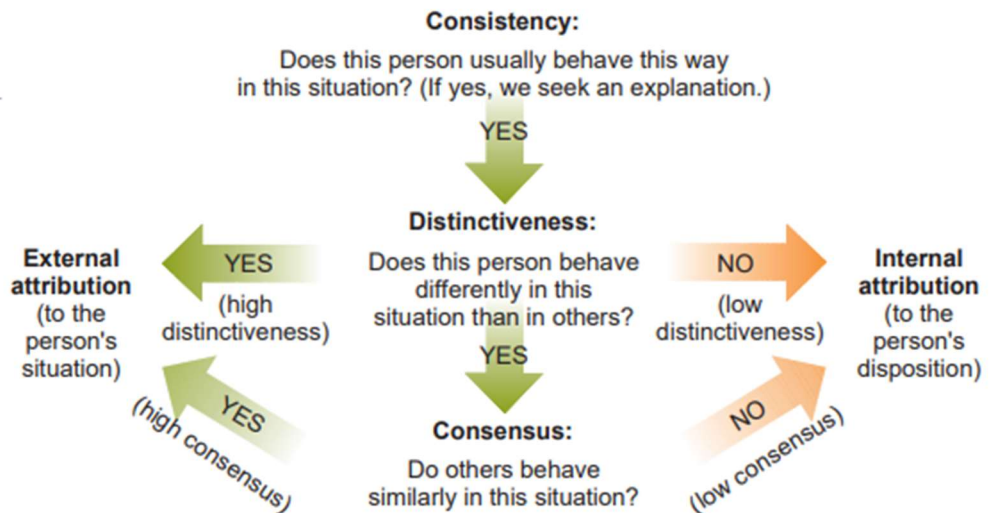
COMMONSENSE ATTRIBUTIONS

As the theory of correspondent inferences suggests, attributions often are rational. Pioneering attribution theorist Harold Kelley (1973) described how we explain behavior by using information about “consistency,” “distinctiveness,” and “consensus” (Figure 3.4).

FIGURE :: 3.4

Harold Kelley's Theory of Attributions

Three factors—consistency, distinctiveness, and consensus—influence whether we attribute someone's behavior to internal or external causes. Try creating your own examples, such as: If Mary and many others criticize Steve (with consensus), and if Mary isn't critical of others (high distinctiveness), then we make an external attribution (it's something about Steve). If Mary alone (low consensus) criticizes Steve, and if she criticizes many other people, too (low distinctiveness), then we are drawn to an internal attribution (it's something about Mary).



Consistency: How consistent is the person's behavior in this situation?

Distinctiveness: How specific is the person's behavior to this particular situation?

Consensus: To what extent do others in this situation behave similarly?

When explaining why Edgar is having trouble with his computer, most people use information concerning consistency (Is Edgar usually unable to get his computer to work?), distinctiveness (Does Edgar have trouble with other computers, or only this one?), and consensus (Do other people have similar problems with this make of computer?). If we learn that Edgar alone consistently has trouble with this and other computers, we likely will attribute the troubles to Edgar, not to defects in this computer.

So our commonsense psychology often explains behavior logically. But Kelley also found that people often discount a contributing cause of behavior if other plausible causes are already known. If we can specify one or two sufficient reasons a student might have done poorly on an exam, we often ignore or discount alternative possibilities (McClure, 1998). Or consider this: Would you guess that people would overestimate or underestimate the frequency of a very famous name, such as “Bush,” in the American population? It surprised me—you, too?—to read Daniel Oppenheimer's (2004) discovery that people under estimate the frequency of hyperfamous names such as Bush relative to an equally common name such as Stevenson. They do so because their familiarity with the name can be attributed to President Bush, which leads them to discount other reasons for their familiarity with “Bush.”

The Fundamental Attribution Error

Social psychology's most important lesson concerns the influence of our social environment. At any moment, our internal state, and therefore what we say and do, depends on the situation as well as on what we bring to the situation. In experiments, a slight difference between two situations sometimes greatly affects how people respond. As a professor, I have seen this when teaching the same subject at both 8:30 a.m. and 7:00 p.m. Silent stares would greet me at 8:30; at

7:00 I had to break up a party. In each situation some individuals were more talkative than others, but the difference between the two situations exceeded the individual differences.

Attribution researchers have found a common problem with our attributions. When explaining someone's behavior, we often underestimate the impact of the situation and overestimate the extent to which it reflects the individual's traits and attitudes. Thus, even knowing the effect of the time of day on classroom conversation, I found it terribly tempting to assume that the people in the 7:00 p.m. class were more extraverted than the "silent types" who came at 8:30 a.m. Likewise, we may infer that people fall because they're clumsy, rather than because they were tripped; that people smile because they're happy rather than faking friendliness; that people speed past us on the highway because they're aggressive rather than late for an important meeting.

fundamental attribution error

The tendency for observers to underestimate situational influences and overestimate dispositional influences upon others' behavior. (Also called *correspondence bias*, because we so often see behavior as corresponding to a disposition.)

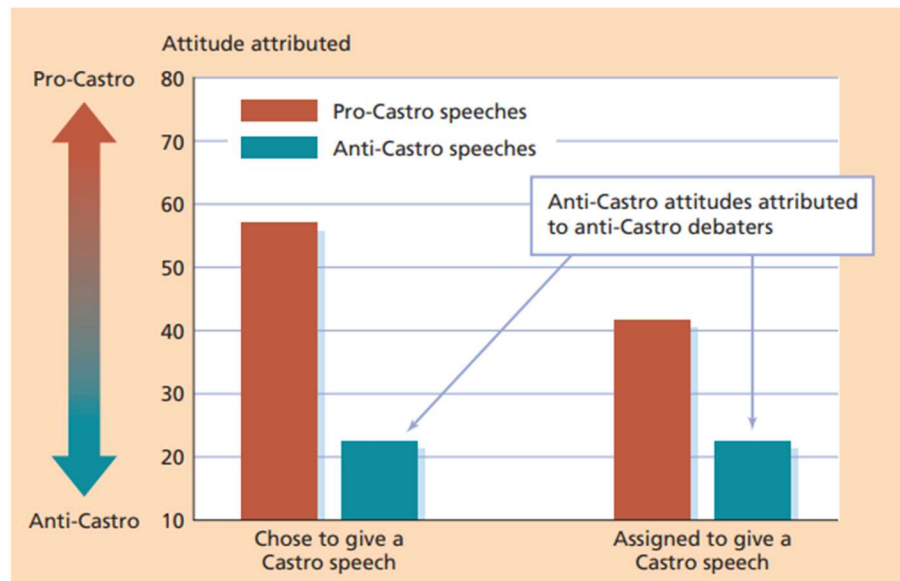
This discounting of the situation, dubbed by Lee Ross (1977) the **fundamental attribution error**, appears in many experiments. In the first such study, Edward Jones and Victor Harris (1967) had Duke University students read debaters' speeches supporting or attacking Cuba's leader, Fidel Castro. When told that the debater chose which position to take, the students logically enough assumed it reflected the person's own attitude. But what happened when the students were told that the debate coach had assigned the position? People who are merely feigning a position write more forceful statements than you'd expect (Allison & others, 1993; Miller & others, 1990). Thus, even knowing that the debater had been told to take a pro- or anti-Castro position did not prevent students from inferring that the debater in fact had the assigned leanings (Figure 3.5). People seemed to think, "Yeah, I know he was assigned that position, but, you know, I think he really believes it."

FIGURE :: 3.5

The Fundamental Attribution Error

When people read a debate speech supporting or attacking Fidel Castro, they attributed corresponding attitudes to the speechwriter, even when the debate coach assigned the writer's position.

Source: Data from Jones & Harris, 1967.



The error is so irresistible that even when people know they are causing someone else's behavior, they still underestimate external influences. If individuals dictate an opinion that someone else must then express, they still tend to see the person as actually holding that opinion (Gilbert & Jones, 1986). If people are asked to be either self-enhancing or self-deprecating during an interview, they are very aware of why they are acting so. But they are unaware of their effect on another person. If Juan acts modestly, his naive partner Bob is likely to exhibit modesty as well. Juan will easily understand his own behavior, but he will think that poor Bob suffers from low self-esteem (Baumeister & others, 1988). In short, we tend to presume that others are the way they act. Observing Cinderella cowering in her oppressive home, people (ignoring the situation) infer that she is meek; dancing with her at the ball, the prince sees a suave and glamorous person.

Lesson 3

The discounting of social constraints was evident in a thought-provoking experiment by Lee Ross and his collaborators (Ross & others, 1977). The experiment re-created Ross's firsthand experience of moving from graduate student to professor. His doctoral oral exam had proved a humbling experience as his apparently brilliant professors quizzed him on topics they specialized in. Six months later, Dr. Ross was himself an examiner, now able to ask penetrating questions on his favorite topics. Ross's hapless student later confessed to feeling exactly as Ross had a half-year before—dissatisfied with his ignorance and impressed with the apparent brilliance of the examiners.

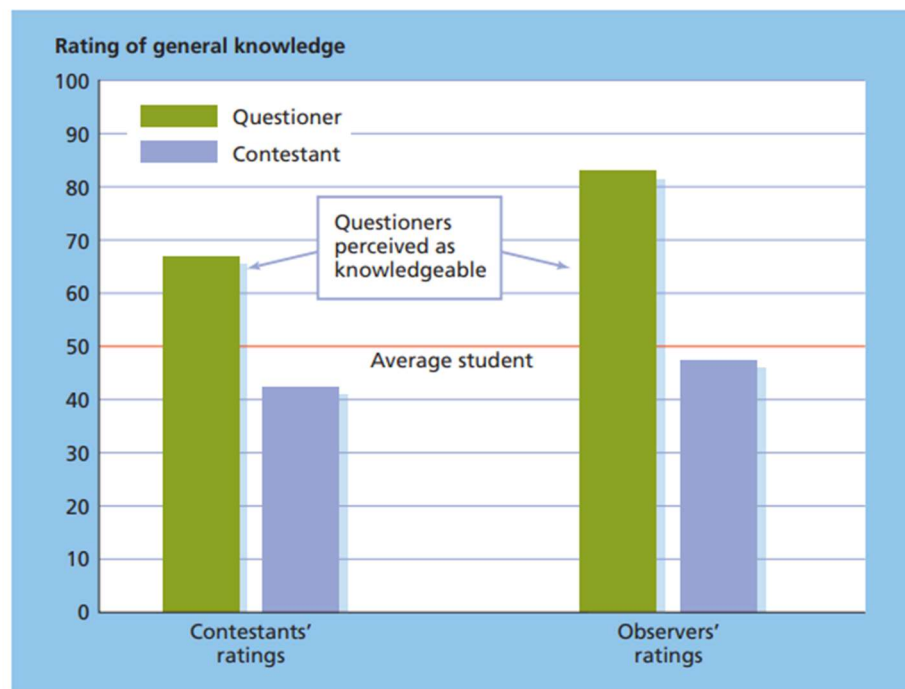
In the experiment, with Teresa Amabile and Julia Steinmetz, Ross set up a simulated quiz game. He randomly assigned some Stanford University students to play the role of questioner, some to play the role of contestant, and others to observe. The researchers invited the questioners to make up difficult questions that would demonstrate their wealth of knowledge. Any one of us can imagine such questions using one's own domain of competence: "Where is Bainbridge Island?" "How did Mary, Queen of Scots, die?" "Which has the longer coastline, Europe or Africa?" If even those few questions have you feeling a little uninformed, then you will appreciate the results of this experiment.

Everyone had to know that the questioner would have the advantage. Yet both contestants and observers (but not the questioners) came to the erroneous conclusion that the questioners really were more knowledgeable than the contestants (Figure 3.6). Follow-up research shows that these misimpressions are hardly a reflection of low social intelligence. If anything, intelligent and socially competent people are more likely to make the attribution error (Block & Funder, 1986).

FIGURE :: 3.6

Both contestants and observers of a simulated quiz game assumed that a person who had been randomly assigned the role of questioner was far more knowledgeable than the contestant. Actually the assigned roles of questioner and contestant simply made the questioner seem more knowledgeable. The failure to appreciate this illustrates the fundamental attribution error.

Source: Data from Ross, Amabile, & Steinmetz, 1977.



In real life, those with social power usually initiate and control conversations, which often leads underlings to overestimate their knowledge and intelligence. Medical doctors, for example, are often presumed to be experts on all sorts of questions unrelated to medicine. Similarly, students often overestimate the brilliance of their teachers. (As in the experiment, teachers are questioners on subjects of their special expertise.) **When some of these students later become teachers, they are usually amazed to discover that teachers are not so brilliant after all.**

To illustrate the fundamental attribution error, most of us need look no further than our own experiences. Determined to make some new friends, Bev plasters a smile on her face and anxiously plunges into a party. Everyone else seems quite relaxed and happy as they laugh and talk with one another. Bev wonders to herself, "Why is everyone always so at ease in groups like this while I'm feeling shy and tense?" Actually, everyone else is feeling nervous, too, and making the same attribution error in assuming that Bev and the others are as they appear - confidently convivial.

WHY DO WE MAKE THE ATTRIBUTION ERROR?

So far we have seen a bias in the way we explain other people's behavior: We often ignore powerful situational determinants. Why do we tend to underestimate the situational determinants of others' behavior but not of our own?

PERSPECTIVE AND SITUATIONAL AWARENESS

Actor versus Observer Perspectives? Attribution theorists pointed out that we observe others from a different perspective than we observe ourselves (Jones, 1976; Jones & Nisbett, 1971). When we act, the environment commands our attention. When we watch another person act, that person occupies the center of our attention and the environment becomes relatively invisible. Auschwitz commandant Rudolph Höss (1959), while acting as a good SS officer "who could not show the slightest trace of emotion," professed inner anguish over his genocidal actions, saying he felt "pity so great that I longed to vanish from the scene." Yet he inferred that his similarly stoic Jewish inmates were uncaring—a "racial characteristic," he presumed—as they led fellow inmates to the gas chambers.

From his analysis of 173 studies, Bertram Malle (2006) concluded that the actor-observer difference is minimal. When our action feels intentional and admirable, we attribute it to our own good reasons, not to the situation. It's only when we behave badly that we're more likely to attribute our behavior to the situation, while someone observing us may spontaneously infer a trait.

The Camera Perspective Bias. In some experiments, people have viewed a videotape of a suspect confessing during a police interview. If they viewed the confession through a camera focused on the suspect, they perceived the confession as genuine. If they viewed it through a camera focused on the detective, they perceived it as more coerced (Lassiter & others, 1986, 2005, 2007). The camera perspective influenced people's guilt judgments even when the judge instructed them not to allow this to happen (Lassiter & others, 2002).

In courtrooms, most confession videotapes focus on the confessor. As we might expect, noted Daniel Lassiter and Kimberly Dudley (1991), such tapes yield a nearly 100 percent conviction rate when played by prosecutors. Aware of this research, reports Lassiter, New Zealand has made it a national policy that police interrogations be filmed with equal focus on the officer and the suspect, such as by filming them with side profiles of both.

Perspectives Change with Time. As the once-visible person recedes in their memory, observers often give more and more credit to the situation. As we saw above in the groundbreaking attribution error experiment by Edward Jones and Victor Harris (1967), immediately after hearing someone argue an assigned position, people assume that's how the person really felt. Jerry Burger and M. L. Palmer (1991) found that a week later they are much more ready to credit the situational constraints. The day after a presidential election, Burger and Julie Pavelich (1994) asked voters why the election turned out as it did. Most attributed the outcome to the candidates' personal traits and positions (the winner from the incumbent party was likable). When they asked other voters the same question a year later, only a third attributed the verdict to the candidates. More people now credited circumstances, such as the country's good mood and the robust economy.

"And in imagination he began to recall the best moments of his pleasant life. . . . But the child who had experienced that happiness existed no longer, it was like a reminiscence of somebody else."

—LEO TOLSTOY, THE DEATH OF IVAN ILYICH, 1886

Let's make this personal: Are you generally quiet, talkative, or does it depend on the situation? "Depends on the situation" is a common answer. But when asked to describe a friend—or to describe what they were like five years ago—people more often ascribe trait descriptions. When recalling our past, we become like observers of someone else, note researchers Emily Pronin and Lee Ross (2006). For most of us, the "old you" is someone other than today's "real you." We regard our distant past selves (and our distant future selves) almost as if they were other people occupying our body.

Self-Awareness. Circumstances can also shift our perspective on ourselves. Seeing ourselves on television redirects our attention to ourselves. Seeing ourselves in a mirror, hearing our tape-recorded voices, having our pictures taken, or filling out biographical questionnaires similarly focuses our attention inward, making us self-conscious instead of situation-conscious. Looking back on ill-fated relationships that once seemed like the unsinkable Titanic, people can more easily see the icebergs (Berscheid, 1999).

Robert Wicklund, Shelley Duval, and their collaborators have explored the effects of **self-awareness** (Duval & Wicklund, 1972; Silvia & Duval, 2001). When our attention focuses upon ourselves, we often attribute responsibility to ourselves. Allan Fenigstein and Charles Carver (1978) demonstrated this by having students imagine themselves in hypothetical situations. Some students were made self-aware by thinking they were hearing their own heartbeats while pondering the situation. Compared with those who thought they were just hearing extraneous noises, the self-aware students saw themselves as more responsible for the imagined outcome.

Some people are typically quite self-conscious. In experiments, people who report themselves as privately self-conscious (who agree with statements such as “I’m generally attentive to my inner feelings”) behave similarly to people whose attention has been self-focused with a mirror (Carver & Scheier, 1978). Thus, people whose attention focuses on themselves—either briefly during an experiment or because they are self-conscious persons—view themselves more as observers typically do; they attribute their behavior more to internal factors and less to the situation.

All these experiments point to a reason for the attribution error: We find causes where we look for them. To see this in your own experience, consider: Would you say your social psychology instructor is a quiet or a talkative person?

My guess is you inferred that he or she is fairly outgoing. But consider: Your attention focuses on your instructor while he or she behaves in a public context that demands speaking. The instructor also observes his or her own behavior in many different situations—in the classroom, in meetings, at home. “Me talkative?” your instructor might say. “Well, it all depends on the situation. When I’m in class or with good friends, I’m rather outgoing. But at conventions and in unfamiliar situations I feel and act rather shy.” ***Because we are acutely aware of how our behavior varies with the situation, we see ourselves as more variable than other people*** (Baxter & Goldberg, 1987; Kammer, 1982; Sande & others, 1988). “Nigel is uptight, Fiona is relaxed. With me it varies.”

CULTURAL DIFFERENCES

Cultures also influence attribution error (Ickes, 1980; Watson, 1982). A Western worldview predisposes people to assume that people, not situations, cause events. Internal explanations are more socially approved (Jellison & Green, 1981). “You can do it!” we are assured by the pop psychology of positive-thinking Western culture. You get what you deserve and deserve what you get.

As children grow up in Western culture, they learn to explain behavior in terms of the other’s personal characteristics (Rholes & others, 1990; Ross, 1981). As a first-grader, one of my sons brought home an example. He unscrambled the words “gate the sleeve caught Tom on his” into “The gate caught Tom on his sleeve.” His teacher, applying the Western cultural assumptions of the curriculum materials, marked that wrong. The “right” answer located the cause within Tom: “Tom caught his sleeve on the gate.”

The fundamental attribution error occurs across varied cultures (Krull & others, 1999). Yet people in Eastern Asian cultures are somewhat more sensitive to the importance of situations. Thus, when aware of the social context, they are less inclined to assume that others’ behavior corresponds to their traits (Choi & others, 1999; Farwell & Weiner, 2000; Masuda & Kitayama, 2004).

Some languages promote external attributions. Instead of “I was late,” Spanish idiom allows one to say, “The clock caused me to be late.” In collectivist cultures, people less often perceive others in terms of personal dispositions (Lee & others,

1996; Zebrowitz-McArthur, 1988). They are less likely to spontaneously interpret a behavior as reflecting an inner trait (Newman, 1993). When told of someone's actions, Hindus in India are less likely than Americans to offer dispositional explanations ("She is kind") and more likely to offer situational explanations ("Her friends were with her") (Miller, 1984).

The fundamental attribution error is *fundamental* because it colors our explanations in basic and important ways. Researchers in Britain, India, Australia, and the United States have found that people's attributions predict their attitudes toward the poor and the unemployed (Furnham, 1982; Pandey & others, 1982; Skitka, 1999; Wagstaff, 1983; Zucker & Weiner, 1993). Those who attribute poverty and unemployment to personal dispositions ("They're just lazy and undeserving") tend to adopt political positions unsympathetic to such people (Figure 3.7). This dispositional attribution ascribes behavior to the person's disposition and traits. Those who make situational attributions ("If you or I were to live with the same overcrowding, poor education, and discrimination, would we be any better off?") tend to adopt political positions that offer more direct support to the poor.

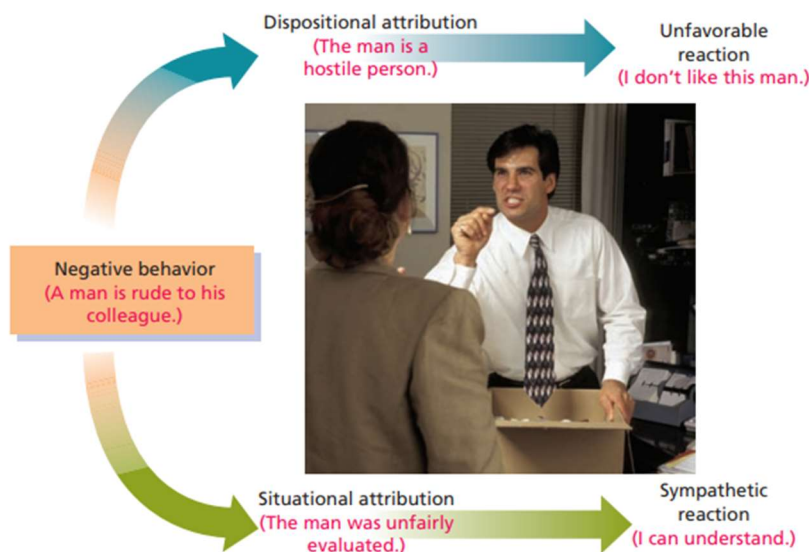


FIGURE :: 3.7

Attributions and Reactions

How we explain someone's negative behavior determines how we feel about it.

Can we benefit from being aware of the attribution error? I once assisted with some interviews for a faculty position. One candidate was interviewed by six of us at once; each of us had the opportunity to ask two or three questions. I came away thinking, "What a stiff, awkward person he is." The second candidate I met privately over coffee, and we immediately discovered we had a close, mutual friend. As we talked, I became increasingly impressed by what a "warm, engaging, stimulating person she is." Only later did I remember the fundamental attribution error and reassess my analysis. I had attributed his stiffness and her warmth to their dispositions; in fact, I later realized, such behavior resulted partly from the difference in their interview situations.

WHY WE STUDY ATTRIBUTION ERRORS

This chapter, like the one before it, explains some foibles and fallacies in our social thinking. Reading about these may make it seem, as one of my students put it, that "social psychologists get their kicks out of playing tricks on people." Actually, the experiments are not designed to demonstrate "what fools these mortals be" (although some of the experiments are rather amusing). Rather, their purpose is to reveal how we think about ourselves and others.

If our capacity for illusion and self-deception is shocking, remember that our modes of thought are generally adaptive. Illusory thinking is often a by-product of our mind's strategies for simplifying complex information. It parallels our perceptual mechanisms, which generally give us useful images of the world but sometimes lead us astray.

A second reason for focusing on thinking biases such as the fundamental attribution error is humanitarian. One of social psychology's "great humanizing messages," note Thomas Gilovich and Richard Eibach (2001), is that **people should not always be blamed for their problems**. "More often than people are willing to acknowledge," they conclude, "failure, disability, and misfortune are . . . the product of real environmental causes."

A third reason for focusing on biases is that we are mostly unaware of them and can benefit from greater awareness. As with other biases, such as the self-serving bias (Chapter 2), people see themselves as less susceptible than others to attribution errors (Pronin, 2008). My hunch is that you will find more surprises, more challenges, and more benefit in an analysis of errors and biases than you would in a string of testimonies to the human capacity for logic and intellectual achievement. That is also why world literature so often portrays pride and other human failings. Social psychology aims to expose us to fallacies in our thinking in the hope that we will become more rational, more in touch with reality. The hope is not in vain: Psychology students explain behavior less simplistically than similarly intelligent natural science students (Fletcher & others, 1986).

EXPECTATIONS OF OUR SOCIAL WORLDS

Having considered how we explain and judge others—efficiently, adaptively, but sometimes erroneously—we conclude this chapter by pondering the effects of our social judgments. Do our social beliefs matter? Do they change reality?

Our social beliefs and judgments do matter. They influence how we feel and act, and by so doing may help generate their own reality. When our ideas lead us to act in ways that produce their apparent confirmation, they have become what sociologist Robert Merton (1948) termed **self-fulfilling prophecies**—beliefs that lead to their own fulfillment. If, led to believe that their bank is about to crash, its customers race to withdraw their money, then their false perceptions may create reality, noted Merton. If people are led to believe that stocks are about to soar, they will indeed. (See "Focus On: The Self-Fulfilling Psychology of the Stock Market.")

focus ON

The Self-Fulfilling Psychology of the Stock Market

On the evening of January 6, 1981, Joseph Granville, a popular Florida investment adviser, wired his clients: "Stock prices will nosedive; sell tomorrow." Word of Granville's advice soon spread, and January 7 became the heaviest day of trading in the previous history of the New York Stock Exchange. All told, stock values lost \$40 billion.

Nearly a half-century ago, John Maynard Keynes likened such stock market psychology to the popular beauty contests then conducted by London newspapers. To win, one had to pick the six faces out of a hundred that were, in turn, chosen most frequently by the other newspaper contestants. Thus, as Keynes wrote, "Each competitor has to pick not those faces which he himself finds prettiest, but those which he thinks likeliest to catch the fancy of the other competitors."

Investors likewise try to pick not the stocks that touch their fancy but the stocks that other investors will favor. The name of the game is predicting others' behavior. As one Wall Street fund manager explained, "You may or may not agree with Granville's view—but that's usually beside the point." If you think his advice will cause others to sell, then you want to sell quickly, before prices

drop more. If you expect others to buy, you buy now to beat the rush.

The self-fulfilling psychology of the stock market worked to an extreme on Monday, October 19, 1987, when the Dow Jones Industrial Average lost 20 percent. Part of what happens during such crashes is that the media and the rumor mill focus on whatever bad news is available to explain them. Once reported, the explanatory news stories further diminish people's expectations, causing declining prices to fall still lower. The process also works in reverse by amplifying good news when stock prices are rising.

In April of 2000, the volatile technology market again demonstrated a self-fulfilling psychology, now called "momentum investing." After two years of eagerly buying stocks (because prices were rising), people started frantically selling them (because prices were falling). Such wild market swings—"irrational exuberance" followed by a crash—are mainly self-generated, noted economist Robert Shiller (2000). In 2008 and 2009, the market psychology headed south again as another bubble burst.

In his well-known studies of *experimenter bias*, Robert Rosenthal (1985, 2006) found that research participants sometimes live up to what they believe experimenters expect of them. In one study, experimenters asked individuals to judge the success of people in various photographs. The experimenters read the same instructions to all their participants and showed them the same photos. Nevertheless, experimenters who expected their participants to see the photographed people as successful obtained higher ratings than did those who expected their participants to see the people as failures. Even more startling—and controversial—are reports that teachers' beliefs about their students similarly serve as self-fulfilling prophecies. If a teacher believes a student is good at math, will the student do well in the class? Let's examine this.

Teacher Expectations and Student Performance

Teachers do have higher expectations for some students than for others. Perhaps you have detected this after having a brother or sister precede you in school, or after receiving a label such as “gifted” or “learning disabled,” or after being tracked with “high-ability” or “average-ability” students. Perhaps conversation in the teachers' lounge sent your reputation ahead of you. Or perhaps your new teacher scrutinized your school file or discovered your family's social status. It's clear that teachers' evaluations correlate with student achievement: Teachers think well of students who do well. That's mostly because teachers accurately perceive their students' abilities and achievements (Jussim, 2005).

But are teachers' evaluations ever a cause as well as a consequence of student performance? One correlational study of 4,300 British schoolchildren by William Crano and Phyllis Mellon (1978) suggested yes. Not only is high performance followed by higher teacher evaluations, but the reverse is true as well.

Could we test this “teacher-expectations effect” experimentally? Pretend we gave a teacher the impression that Dana, Sally, Todd, and Manuel—four randomly selected students—are unusually capable. Will the teacher give special treatment to these four and elicit superior performance from them? In a now-famous experiment, Rosenthal and Lenore Jacobson (1968) reported precisely that. Randomly selected children in a San Francisco elementary school who were said (on the basis of a fictitious test) to be on the verge of a dramatic intellectual spurt did then spurt ahead in IQ score.

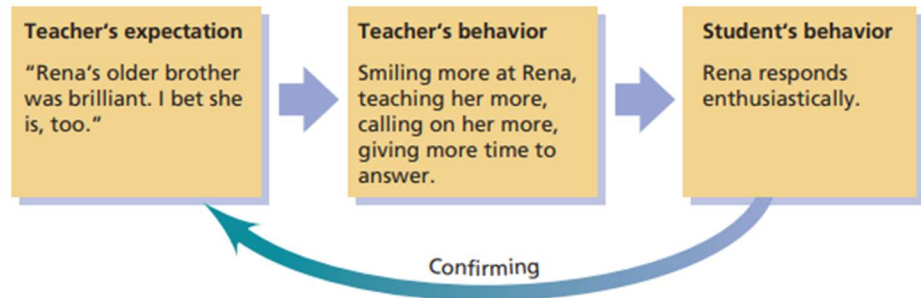
That dramatic result seemed to suggest that the school problems of “disadvantaged” children might reflect their teachers' low expectations. The findings were soon publicized in the national media as well as in many college textbooks. However, further analysis—which was not as highly publicized—revealed the teacher-expectations effect to be not as powerful and reliable as this initial study had led many people to believe (Spitz, 1999). By Rosenthal's own count, in only about 4 in 10 of the nearly 500 published experiments did expectations significantly affect performance (Rosenthal, 1991, 2002). Low expectations do not doom a capable child, nor do high expectations magically transform a slow learner into a valedictorian. Human nature is not so pliable.

High expectations do, however, seem to boost low achievers, for whom a teacher's positive attitude may be a hope-giving breath of fresh air (Madon & others, 1997). How are such expectations transmitted? Rosenthal and other investigators report that teachers look, smile, and nod more at “high-potential students.” Teachers also may teach more to their “gifted” students, set higher goals for them, call on them more, and give them more time to answer (Cooper, 1983; Harris & Rosenthal, 1985, 1986; Jussim, 1986).

In one study, Elisha Babad, Frank Bernieri, and Rosenthal (1991) videotaped teachers talking to, or about, unseen students for whom they held high or low expectations. A random 10-second clip of either the teacher's voice or the teacher's face was enough to tell viewers—both children and adults—whether this was a good or a poor student and how much the teacher liked the student. (You read that right: 10 seconds.) Although teachers may think they can conceal their feelings and behave impartially toward the class, students are acutely sensitive to teachers' facial expressions and body movements (Figure 3.8).

FIGURE :: 3.8**Self-Fulfilling Prophecies**

Teacher expectations can become self-fulfilling prophecies. But for the most part, teachers' expectations accurately reflect reality (Jussim & Harber, 2005).



Reading the experiments on teacher expectations makes me wonder about the effect of students' expectations upon their teachers. You no doubt begin many of your courses having heard "Professor Smith is interesting" and "Professor Jones is a bore." Robert Feldman and Thomas Prohaska (1979; Feldman & Theiss, 1982) found that such expectations can affect both student and teacher. Students in a learning experiment who expected to be taught by an excellent teacher perceived their teacher (who was unaware of their expectations) as more competent and interesting than did students with low expectations. Furthermore, the students actually learned more. In a later experiment, women who were led to expect their male instructor to be sexist had a less positive experience with him, performed worse, and rated him as less competent than did women not given the sexist expectation (Adams & others, 2006).

Were these results due entirely to the students' perceptions, or also to a self-fulfilling prophecy that affected the teacher? In a follow-up experiment, Feldman and Prohaska videotaped teachers and had observers rate their performances. Teachers were judged most capable when assigned a student who nonverbally conveyed positive expectations.

To see whether such effects might also occur in actual classrooms, a research team led by David Jamieson (1987) experimented with four Ontario high school classes taught by a newly transferred teacher. During individual interviews, they told students in two of the classes that both other students and the research team rated the teacher very highly. Compared with the control classes, students who were given positive expectations paid better attention during class. At the end of the teaching unit, they also got better grades and rated the teacher as clearer in her teaching. The attitudes that a class has toward its teacher are as important, it seems, as the teacher's attitude toward the students.

Getting from Others What We Expect

So the expectations of experimenters and teachers, though usually reasonably accurate, occasionally act as self-fulfilling prophecies. How widespread are self-fulfilling prophecies? Do we get from others what we expect of them? Studies show that self-fulfilling prophecies also operate in work settings (with managers who have high or low expectations), in courtrooms (as judges instruct juries), and in simulated police contexts (as interrogators with guilty or innocent expectations interrogate and pressure suspects) (Kassin & others, 2003; Rosenthal, 2003, 2006).

Do self-fulfilling prophecies color our personal relationships? There are times when negative expectations of someone lead us to be extra nice to that person, which induces him or her to be nice in return—thus disconfirming our expectations. But a more common finding in studies of social interaction is that, yes, we do to some extent get what we expect (Olson & others, 1996).

In laboratory games, hostility nearly always begets hostility: People who perceive their opponents as noncooperative will readily induce them to be noncooperative (Kelley & Stahelski, 1970). Each party's perception of the other as aggressive, resentful, and vindictive induces the other to display those behaviors in self-defense, thus creating a vicious self-perpetuating circle. In another experiment, people anticipated interacting with another person of a different race. When led to expect that the person disliked interacting with someone of their race, they felt more anger and displayed more hostility toward the person (Butz & Plant, 2006). Likewise, whether I expect my wife to be in a bad mood or in a loving mood may affect how I relate to her, thereby inducing her to confirm my belief.

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So, do intimate relationships prosper when partners idealize each other? Are positive illusions of the other's virtues self-fulfilling? Or are they more often self-defeating, by creating high expectations that can't be met? Among University of Waterloo dating couples followed by Sandra Murray and her associates (1996a, 1996b, 2000), positive ideals of one's partner were good omens. Idealization helped buffer conflict, bolster satisfaction, and turn self-perceived frogs into princes or princesses. When someone loves and admires us, it helps us become more the person he or she imagines us to be.

When dating couples deal with conflicts, hopeful optimists and their partners tend to perceive each other as engaging constructively. Compared to those with more pessimistic expectations, they then feel more supported and more satisfied with the outcome (Srivastava & others, 2006). Among married couples, too, those who worry that their partner doesn't love and accept them interpret slight hurts as rejections, which motivates them to devalue the partner and distance themselves. Those who presume their partner's love and acceptance respond less defensively, read less into stressful events, and treat the partner better (Murray & others, 2003). ***Love helps create its presumed reality.***

Several experiments conducted by Mark Snyder (1984) at the University of Minnesota show how, once formed, erroneous beliefs about the social world can induce others to confirm those beliefs, a phenomenon called ***behavioral confirmation***. In a classic study, Snyder, Elizabeth Tanke, and Ellen Berscheid (1977) had male students talk on the telephone with women they thought (from having been shown a picture) were either attractive or unattractive. Analysis of just the women's comments during the conversations revealed that the supposedly attractive women spoke more warmly than the supposedly unattractive women. The men's erroneous beliefs had become a self-fulfilling prophecy by leading them to act in a way that influenced the women to fulfill the men's stereotype that beautiful people are desirable people.

Behavioral confirmation also occurs as people interact with partners holding mistaken beliefs. People who are believed lonely behave less sociably (Rotenberg & others, 2002). Men who are believed sexist behave less favorably toward women (Pinel, 2002). Job interviewees who are believed to be warm behave more warmly.

Imagine yourself as one of the 60 young men or 60 young women in an experiment by Robert Ridge and Jeffrey Reber (2002). Each man is to interview one of the women to assess her suitability for a teaching assistant position. Before doing so, he is told either that she feels attracted to him (based on his answers to a biographical questionnaire) or not attracted. (Imagine being told that someone you were about to meet reported considerable interest in getting to know you and in dating you, or none whatsoever.) The result was behavioral confirmation: Applicants believed to feel an attraction exhibited more flirtatiousness (and without being aware of doing so). Ridge and Reber believe that this process, like the misattribution phenomenon we discussed earlier, may be one of the roots of sexual harassment. If a woman's behavior seems to confirm a man's beliefs, he may then escalate his overtures until they become sufficiently overt for the woman to recognize and interpret them as inappropriate or harassing.

Expectations influence children's behavior, too. After observing the amount of litter in three classrooms, Richard Miller and his colleagues (1975) had the teacher and others repeatedly tell one class that they should be neat and tidy. This persuasion increased the amount of litter placed in wastebaskets from 15 to 45 percent, but only temporarily. Another class, which also had been placing only 15 percent of its litter in wastebaskets, was repeatedly congratulated for being so neat and tidy. After eight days of hearing this, and still two weeks later, these children were fulfilling the expectation by putting more than 80 percent of their litter in wastebaskets. Tell children they are hardworking and kind (rather than lazy and mean), and they may live up to their labels.

These experiments help us understand how social beliefs, such as stereotypes about people with disabilities or about people of a particular race or sex, may be self-confirming. How others treat us reflects how we and others have treated them.

A note of caution: As with every social phenomenon, the tendency to confirm others' expectations has its limits. Expectations often predict behavior simply because they are accurate (Jussim, 2005).

CONCLUSION

Social cognition studies reveal that our information-processing powers are impressive for their efficiency and adaptiveness ("in apprehension how like a god!" exclaimed Shakespeare's Hamlet). Yet we are also vulnerable to predictable errors and misjudgments ("headpiece filled with straw," said T. S. Eliot). What practical lessons, and what insights into human nature, can we take home from this research?

We have reviewed reasons why people sometimes form false beliefs. We cannot easily dismiss these experiments: Most of their participants were intelligent people, often students at leading universities. Moreover, **people's intelligence scores are uncorrelated with their vulnerability to many different thinking biases** (Stanovich & West, 2008). One can be very smart and exhibit seriously bad judgment.

Trying hard also doesn't eliminate thinking biases. These predictable distortions and biases occurred even when payment for right answers motivated people to think optimally. As one researcher concluded, the illusions "have a persistent quality not unlike that of perceptual illusions" (Slovic, 1972).

Research in cognitive social psychology thus mirrors the mixed review given humanity in literature, philosophy, and religion. Many research psychologists have spent lifetimes exploring the awesome capacities of the human mind. We are smart enough to have cracked our own genetic code, to have invented talking computers, to have sent people to the moon. Three cheers for human reason.

Well, two cheers—because the mind's premium on efficient judgment makes our intuition more vulnerable to misjudgment than we suspect. With remarkable ease, we form and sustain false beliefs. Led by our preconceptions, feeling overconfident, persuaded by vivid anecdotes, perceiving correlations and control even where none may exist, we construct our social beliefs and then influence others to confirm them. "The naked intellect," observed novelist Madeleine L'Engle, "is an extraordinarily inaccurate instrument."

But have these experiments just been intellectual tricks played on hapless participants, thus making them look worse than they are? Richard Nisbett and Lee Ross (1980) contended that, if anything, laboratory procedures overestimate our intuitive powers. The experiments usually present people with clear evidence and warn them that their reasoning ability is being tested. Seldom does real life say to us: "Here is some evidence. Now put on your intellectual Sunday best and answer these questions."

Often our everyday failings are inconsequential, but not always so. False impressions, interpretations, and beliefs can produce serious consequences. Even small biases can have profound social effects when we are making important social judgments: Why are so many people homeless? unhappy? homicidal? Does my friend love me or my money? Cognitive biases even creep into sophisticated scientific thinking. Human nature has hardly changed in the 3,000 years since the Old Testament psalmist noted that "no one can see his own errors."

Is this too cynical? Leonard Martin and Ralph Erber (2005) invite us to imagine that an intelligent being swooped down and begged for information that would help it understand the human species. When you hand it this social psychology text, the alien says "thank you" and zooms back off into space. After (I'd like to presume) resolving your remorse over giving up this book, how would you feel about having offered social psychology's analysis? Joachim Krueger and David Funder (2003a, 2003b) wouldn't feel too good. Social psychology's preoccupation with human foibles needs balancing with "a more positive view of human nature," they argue.

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Fellow social psychologist Lee Jussim (2005) agrees, adding, “Despite the oft-demonstrated existence of a slew of logical flaws and systematic biases in lay judgment and social perception, such as the fundamental attribution error, false consensus, over-reliance on imperfect heuristics, self-serving biases, etc., people’s perceptions of one another are surprisingly (though rarely perfectly) accurate.” The elegant analyses of the imperfections of our thinking are themselves a tribute to human wisdom. Were one to argue that all human thought is illusory, the assertion would be self-refuting, for it, too, would be but an illusion. It would be logically equivalent to contending “All generalizations are false, including this one.”

As medical science assumes that any given body organ serves a function, so behavioral scientists find it useful to assume that our modes of thought and behavior are adaptive (Funder, 1987; Kruglanski & Ajzen, 1983; Swann, 1984). The rules of thought that produce false beliefs and striking deficiencies in our statistical intuition usually serve us well. Frequently, the errors are a by-product of our mental shortcuts that simplify the complex information we receive.

Nobel laureate psychologist Herbert Simon (1957) was among the modern researchers who first described the bounds of human reason. Simon contends that to cope with reality, we simplify it. Consider the complexity of a chess game: The number of possible games is greater than the number of particles in the universe. How do we cope? We adopt some simplifying rules—heuristics. These heuristics sometimes lead us to defeat. But they do enable us to make efficient snap judgments.

Illusory thinking can likewise spring from useful heuristics that aid our survival. In many ways, heuristics “make us smart” (Gigerenzer, 2007). The belief in our power to control events helps maintain hope and effort. If things are sometimes subject to control and sometimes not, we maximize our outcomes by positive thinking. Optimism pays dividends. We might even say that our beliefs are like scientific theories—sometimes in error yet useful as generalizations. As social psychologist Susan Fiske (1992) says, “Thinking is for doing.”

As we constantly seek to improve our theories, might we not also work to reduce errors in our social thinking? In school, math teachers teach, teach, teach until the mind is finally trained to process numerical information accurately and automatically. We assume that such ability does not come naturally; otherwise, why bother with the years of training? Research psychologist Robyn Dawes (1980)—who was dismayed that “study after study has shown [that] people have very limited abilities to process information on a conscious level, particularly social information”—suggested that we should also teach, teach, teach how to process social information.

Richard Nisbett and Lee Ross (1980) believe that education could indeed reduce our vulnerability to certain types of error. They offer the following recommendations:

- Train people to recognize likely sources of error in their own social intuition.
- Set up statistics courses geared to everyday problems of logic and social judgment. Given such training, people do in fact reason better about everyday events (Lehman & others, 1988; Nisbett & others, 1987).
- Make such teaching more effective by illustrating it richly with concrete, vivid anecdotes and examples from everyday life.
- Teach memorable and useful slogans, such as “It’s an empirical question,” “Which hat did you draw that sample out of?” or “You can lie with statistics, but a well-chosen example does the job better.”

Source:

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